

Designing a Crop Insurance Scheme for Paddy Cultivation in India

Dr. Bappaditya Mukhopadhyay and Samya Mukherjee

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Abstract

This study presents a comprehensive hybrid approach that integrates statistical, machine learning, and financial modeling techniques to forecast rainfall, rice yield, and rice prices across major Indian states and to design actuarially sound crop insurance schemes. We leverage time-series techniques such as Exponentially Weighted Moving Average (EWMA) to smooth and forecast rainfall characteristics (mean and standard deviation), ARIMA to model and predict mandi-level rice prices, and ARIMAX and Long Short-Term Memory (LSTM) neural networks to forecast rice yield using rainfall and area sown as inputs.

To capture the economic impact of crop failures and climate shocks, we embed a financial derivative-like structure in the insurance model, where the farmer's payout is computed using a piecewise payoff function based on the forecasted yield Y relative to two thresholds X_1 and X_2 . Specifically, the payout function is defined as:

- Full payout X_1 when $Y < X_1$
- Partial payout $X_2 - Y$ when $X_1 \leq Y < X_2$
- No payout when $Y \geq X_2$

This approach mirrors structured financial options and allows for nuanced compensation mechanisms depending on the severity of yield loss. The premium is then computed as the discounted expected value of this payout using actuarial principles. We apply the framework to ten major rice-producing states in India using data from 1966 to 2023, and our results demonstrate that this integrated framework provides accurate forecasts and fair premium calculations, supporting both farmers and insurers in risk management.

Keywords: Crop Insurance, Rainfall Forecasting, EWMA, ARIMA, ARIMAX, LSTM, Yield Prediction, Price Forecasting, Financial Derivatives, Climate Risk, Indian Agriculture, Risk Management, Insurance Premium Estimation

Introduction

Agriculture has historically been the backbone of the Indian economy, employing over 40% of the workforce and contributing significantly to rural livelihoods. Despite its critical role, Indian agriculture remains exceptionally vulnerable to climatic and market-related risks. This vulnerability is particularly evident in rain-fed agriculture, which constitutes a major share of cultivated land and is predominantly dependent on the monsoon season.

Among all Kharif crops, rice holds a central place in the Indian food security framework. However, rice production is inherently sensitive to deviations in monsoonal rainfall and hence serves as an ideal proxy for understanding climate-induced agricultural risk.

Rainfall in India exhibits significant spatial and temporal variability. Uneven rainfall distribution, delayed onset, early withdrawal, or extreme rainfall events (droughts or floods) can have a disproportionately large impact on crop yields. Furthermore, climate change is expected to exacerbate these patterns by increasing the frequency and intensity of extreme weather events. Alongside climate variability, agricultural producers also face considerable price uncertainty. Rice prices vary across markets and years due to supply-demand imbalances, input cost changes, global trade dynamics, and policy interventions such as Minimum Support Prices (MSP).

The dual uncertainty arising from climate variability and price volatility creates systemic risk for farmers—especially smallholders—who often lack access to credit, savings, or formal risk mitigation tools. Conventional crop insurance schemes such as the Pradhan Mantri Fasal Bima Yojana (PMFBY) aim to cushion these shocks by providing financial compensation in the event of yield loss. However, these schemes suffer from multiple design and implementation inefficiencies:

- **Delayed compensation:** Due to the reliance on manual crop-cutting experiments (CCEs), payouts often arrive months after losses are incurred.
- **Moral hazard and adverse selection:** Since farmers are not directly involved in loss estimation, there is limited incentive for accurate reporting or risk-reducing behaviour.
- **Basis risk:** The spatial aggregation of yield data often leads to situations where individual farmers suffer losses but receive no payout due to district-level averages.
- **Low actuarial fairness:** Premiums are often not aligned with the actual risk profile of farmers, leading to poor uptake and sustainability of the program.

Against this backdrop, the current study proposes a novel, data-driven approach to design a transparent, timely, and statistically grounded crop insurance mechanism. The core premise is to forecast the probability of yield shortfall and monetize that risk using financial option theory. Our methodology integrates advanced time series models, statistical estimation, and financial engineering to create a predictive and actuarially fair insurance premium system.

The major components of our proposed framework are as follows:

1. **Rainfall Modeling using EWMA:** Rainfall distribution parameters (mean and standard deviation) during the June–September monsoon season are modeled using the Exponentially Weighted Moving Average (EWMA) method. This approach captures recent shifts in climate patterns while retaining long-term stability, thereby enabling the construction of dynamic rainfall risk indices at the district level.
2. **Rice Price Forecasting via ARIMA:** Mandi-level rice prices are forecasted using Autoregressive Integrated Moving Average (ARIMA) models. This enables a forward-looking estimation of the economic value of yield, which is crucial for calculating loss-based payouts.

3. **Rice Yield Prediction via ARIMAX and LSTM:** To incorporate the influence of rainfall on crop yields, we use two complementary approaches—ARIMAX (which includes rainfall as an exogenous regressor) and Long Short-Term Memory (LSTM) neural networks, which capture nonlinear temporal relationships between climate variables and yield outcomes.
4. **Premium Estimation using Option Theory:** Drawing from the Black-Scholes framework, we treat the predicted yield as the underlying asset and define a bounded payout interval using a symmetrical volatility band (typically $\pm 0.5\sigma$). The expected loss is computed using the truncated distribution of yields within this band, and the final premium is obtained by discounting this expected loss to the present using a risk-free rate.

The payoff structure mimics that of a European straddle option, ensuring that farmers are compensated for both moderate and extreme deviations from the expected yield. This method also allows for closed-form analytical computation of premiums and facilitates rapid rollout using real-time data. By aligning insurance premiums with statistical risk, this approach improves actuarial fairness and reduces over-reliance on ex-post damage verification.

Our empirical analysis focuses on ten major rice-producing states across India—Bihar, Chhattisgarh, Himachal Pradesh, Jharkhand, Kerala, Odisha, Punjab, Tamil Nadu, Telangana, and Uttarakhand—using historical panel data from 1966 to 2023. We validate the robustness of our models using both in-sample diagnostics and out-of-sample predictions. The resulting premium estimates are benchmarked against realized rainfall and yield deviations to assess predictive accuracy and fairness.

This research lies at the intersection of agricultural economics, climate risk modeling, and financial mathematics. It offers both methodological innovation and practical policy implications. The proposed framework can aid policymakers, insurance companies, and agri-tech platforms in designing scalable, reliable, and farmer-centric insurance solutions. By leveraging data science and financial theory, this study contributes to building climate resilience and income security for millions of farmers in India.

1 Literature Review

The concept of weather-index-based crop insurance has been a subject of growing interest in agricultural economics, risk management, and development finance. In developing economies like India—where over half the population relies on agriculture for their livelihood and farming is largely rain-fed—the need for effective and scalable insurance mechanisms is paramount. Traditional indemnity-based crop insurance schemes have long suffered from operational inefficiencies, such as delays in claims processing, inaccurate yield measurement, and susceptibility to moral hazard. Against this backdrop, index-based insurance—especially those using weather parameters such as rainfall or temperature—has been proposed as a more objective, timely, and cost-effective alternative.

One of the foundational studies in this space was conducted by Giné, Townsend, and Vickery (2009), who analyzed rainfall-indexed insurance in Andhra Pradesh, India. Their empirical investigation revealed that although the insurance product had the potential to reduce income volatility for farmers, uptake remained low. Key reasons included the presence of *basis risk*—the risk that the index does not correlate well with the actual losses

experienced by the farmer—as well as limited financial literacy and trust in the insurance mechanism. This highlighted the importance of designing localized and statistically robust indices for effective risk transfer.

Manuamorn (2007) explored the broader applicability of microinsurance solutions in Asia, focusing on the feasibility of weather insurance for smallholder farmers. She emphasized that technological advancements in weather monitoring, geospatial mapping, and actuarial modeling could enable scalable, affordable insurance solutions. However, she also pointed out implementation bottlenecks, including regulatory challenges, limited data granularity, and a lack of institutional infrastructure to support rapid claim disbursements. Her work underscored the need for integrating technological innovation with policy support to operationalize such models.

The modeling of weather variables and agricultural outputs is another crucial aspect of the literature. Rainfall modeling in particular plays a central role in the design of weather-index insurance. Several authors have advocated for the use of **EWMA (Exponentially Weighted Moving Average)** techniques to model mean rainfall and its variability over time. Unlike traditional moving averages, the EWMA assigns exponentially decreasing weights to older observations, which allows it to be more responsive to recent climate shifts while maintaining a long-term perspective. This property is particularly important in the context of climate change, where weather patterns are evolving in unpredictable ways. Studies have shown that EWMA-based volatility estimates often outperform fixed-window standard deviation estimates in capturing the short-term risk of rainfall variability.

Parallel to weather modeling is the modeling of agricultural prices. Price volatility is a significant concern for farmers, as market prices fluctuate due to a complex set of factors including supply-demand mismatches, international trade policies, and changing input costs. **ARIMA (AutoRegressive Integrated Moving Average)** models, as introduced by Box and Jenkins (1976), have been widely used for price forecasting. These models are particularly attractive in agricultural economics due to their simplicity, interpretability, and relatively low data requirements. Several studies have successfully applied ARIMA models to forecast rice and wheat prices in India and Southeast Asia. However, the predictive power of ARIMA models often diminishes in the presence of structural breaks or high volatility, necessitating hybrid approaches or model enhancements.

To improve upon univariate forecasting, researchers have developed models that incorporate exogenous predictors, such as rainfall or fertilizer use. The **ARIMAX (ARIMA with exogenous variables)** model extends the ARIMA framework by allowing external inputs to influence the dynamics of the time series. In agricultural yield forecasting, ARIMAX models have shown promise in capturing the causal impact of climate variables on output. For instance, studies have found significant relationships between monsoon rainfall and rice yield, which can be effectively modeled using ARIMAX when reliable climate data is available.

Despite these advancements, traditional linear models like ARIMA and ARIMAX are often limited in capturing the non-linear and high-dimensional interactions that govern agricultural systems. Recent developments in machine learning and deep learning offer promising alternatives. In particular, **Long Short-Term Memory (LSTM)** neural networks—a specialized form of recurrent neural networks (RNNs)—have demonstrated superior performance in time series forecasting tasks involving long-term dependencies. LSTM models are capable of capturing non-linear, lagged effects of exogenous variables such as temperature, rainfall, and soil moisture on crop yields. A growing body of liter-

ature documents the application of LSTM in agriculture, ranging from yield forecasting and drought prediction to crop classification using satellite imagery. For instance, research by Khaki and Wang (2019) applied LSTM networks to corn yield prediction in the United States, achieving significantly higher accuracy compared to linear models.

From the insurance pricing perspective, several works have explored the application of financial engineering tools to agriculture. The pioneering work by Black and Scholes (1973) introduced a closed-form pricing formula for European call and put options, which has since found applications beyond financial markets. In the context of agriculture, option-based frameworks can be used to model insurance payoffs as contingent claims on yield or rainfall. For example, a rainfall-based insurance contract may be structured analogously to a put option, where the payout increases as rainfall falls below a certain threshold.

Clarke and Mahul (2012), in their World Bank policy paper, proposed a weather-index insurance framework that integrates financial option theory to ensure pricing transparency and consistency. Their framework relies on actuarial estimates of expected losses based on historical weather data and calibrates premium levels accordingly. By framing insurance as a structured financial derivative, this approach enables closed-form solutions, easier hedging in reinsurance markets, and reduced reliance on field-level data collection.

Building on these strands of literature, our study proposes a unified modeling framework that integrates classical statistical tools (EWMA, ARIMA, ARIMAX), modern deep learning techniques (LSTM), and a financial option-theoretic approach to insurance premium design. While prior studies have typically focused on one component—such as yield modeling, price forecasting, or index construction—we aim to create an end-to-end pipeline that incorporates climate risk modeling, economic valuation, and financial risk transfer in a single framework.

Furthermore, most of the existing weather insurance products are limited to pilot programs or small-scale implementations due to the lack of robust actuarial pricing and spatially granular weather-yield models. By leveraging decades of historical panel data, including district-wise rainfall and yield records from 1966 to 2023, our framework aims to provide a scalable solution applicable across agro-climatic zones in India. The addition of machine learning elements allows for adaptive modeling under changing climate regimes, while the financial pricing model ensures fairness, interpretability, and regulatory compliance.

In summary, this research contributes to the literature by:

- Combining statistical and machine learning methods for multi-stage forecasting (rainfall, yield, price).
- Proposing an actuarially sound, analytically tractable premium estimation method based on option theory.
- Demonstrating empirical applicability across ten major rice-producing states using long-term historical data.
- Offering policy-relevant insights for designing next-generation, data-driven crop insurance systems.

This multi-disciplinary approach aims not only to improve predictive accuracy but also to enhance trust, scalability, and fairness in agricultural insurance mechanisms—a critical need in the era of climate change and increasing farmer distress.

2 Data Description

This research employs diverse datasets spanning historical yield, rainfall, and mandi prices, combined to forecast rice yield and price at the state level and to design rainfall-based crop insurance for Indian farmers. The integrated datasets enable robust statistical and machine learning modeling, focusing on the Kharif season and rice production.

2.1 ICRISAT District-Level Agricultural Dataset (1966–2015)

The International Crops Research Institute for the Semi-Arid Tropics (ICRISAT) provides one of the most comprehensive and granular agricultural datasets for India. The dataset spans a 50-year period from 1966 to 2015 and is curated to support long-term agricultural research, policy analysis, and climate impact modeling. It aggregates data at the **district level**, making it highly suitable for spatially disaggregated statistical modeling. This level of granularity is critical in a country like India, where agro-climatic conditions, rainfall patterns, and cropping systems vary widely across regions.

Scope and Structure of the Dataset: The ICRISAT dataset contains annual observations for over 400 districts across India. Each row in the dataset represents the agricultural record of a specific district in a specific year. The dataset includes variables spanning geographical identifiers, crop statistics, land use, and production data. For the purpose of this study, we extracted and focused on the following key components:

- **Geographic and Temporal Identifiers:**

- **Dist Code:** A unique numerical identifier assigned to each district.
- **Year:** The agricultural year corresponding to the data record.
- **State Code:** A numerical identifier for the corresponding state.
- **State Name, District Name:** Textual labels providing easy identification and cross-verification.

These fields facilitate multi-year, multi-state analyses and allow merging with external datasets, such as rainfall or price data.

- **Crop-wise Agricultural Statistics:** The dataset provides detailed production data for a wide variety of crops, both food and cash crops. These include:

- RICE, WHEAT, MAIZE, MILLETS, SORGHUM (cereals)
- OILSEEDS, COTTON, SUGARCANE (commercial crops)
- FRUITS, VEGETABLES (horticulture)

Each of these crops has associated statistics recorded under three main categories:

1. **Area (in 1000 hectares)** – The total land area under cultivation for the crop in that district-year.
2. **Production (in 1000 metric tons)** – The gross output harvested.
3. **Yield (in kg/ha)** – Derived as $\text{Production} \div \text{Area}$, this is a key measure of agricultural productivity and efficiency.

Focus on Rice: While the dataset includes data for more than ten crops, our study exclusively focuses on rice due to the following reasons:

- Rice is the most important Kharif crop in India, accounting for a major share of total foodgrain production.
- It is highly sensitive to monsoon rainfall, making it an ideal candidate for climate-based risk modeling.
- Rice production exhibits substantial spatial and temporal variability, offering a rich dataset for machine learning and time series modeling.
- Policy relevance: Several government insurance schemes, procurement policies, and minimum support prices are directly linked to rice.

Preprocessing and Cleaning: Before modeling, the data underwent extensive pre-processing:

- Missing values for area or production were either imputed using temporal interpolation or removed based on consistency checks.
- Yield values were recalculated where inconsistencies in reported figures were detected.
- District boundaries were harmonized with modern census codes to facilitate integration with rainfall data and remote sensing products.
- Years with anomalous yield data due to pest attacks, floods, or droughts were flagged using external metadata for validation purposes.

Use in Modeling: The final rice yield time series (in kg/ha) for each district was used as the dependent variable in our forecasting models. These values were matched with rainfall statistics from the Indian Meteorological Department (IMD) and market price data from Agmarknet. Together, they formed the basis of a three-pillar modeling system:

- Predicting rainfall (mean and standard deviation) via EWMA models.
- Predicting rice prices via ARIMA time series models.
- Predicting rice yield via ARIMAX (with exogenous rainfall input) and LSTM networks.

Conclusion: The ICRISAT dataset served as the backbone of our study, providing a long-term, high-resolution view of agricultural dynamics across India. Its historical depth and spatial richness made it possible to train predictive models with a high degree of granularity, enabling fair and localized insurance premium estimation.

2.2 IMD Monthly Rainfall Dataset (1901–2015)

The Indian Meteorological Department (IMD) maintains one of the most authoritative and comprehensive meteorological databases in India. Among its various datasets, the monthly rainfall statistics from 1901 to 2015 serve as a critical resource for long-term climate modeling, agricultural planning, and drought assessment. The dataset is structured at the level of **meteorological subdivisions**, which are defined geographical zones comprising one or more districts with broadly similar climatic conditions. As of the dataset’s structure, there are 36 such subdivisions covering the entire country.

Structure of the Dataset: The IMD Monthly Rainfall Dataset records precipitation in millimeters (mm) and is arranged in a tabular format with the following key fields:

- **SUBDIVISION:** The meteorological region (e.g., “East Uttar Pradesh”, “South Interior Karnataka”, “Kerala”) to which the data pertains.
- **YEAR:** The calendar year corresponding to the rainfall record.
- **Monthly Rainfall (JAN to DEC):** Twelve columns record monthly rainfall in millimeters (mm), providing a high-resolution view of intra-annual variability.
- **Seasonal Aggregates:** Precomputed seasonal totals for key agricultural periods:
 - **Jan–Feb:** Winter season
 - **Mar–May:** Pre-monsoon season
 - **Jun–Sep:** Southwest Monsoon (Kharif season)
 - **Oct–Dec:** Post-monsoon (Rabi) season
 - **ANNUAL:** Total rainfall over all 12 months

This structure makes the dataset suitable for both seasonal trend analysis and event-based climate modeling (e.g., drought years, excess rainfall events, etc.).

Focus on June–September (Kharif) Rainfall: For this study, we primarily focused on the **June to September** rainfall totals, which correspond to the **Southwest Monsoon** period in India. This season is of paramount importance to Indian agriculture as it accounts for over 70% of the country’s total annual precipitation. The Kharif cropping season—including rice, maize, millet, and pulses—is heavily reliant on timely and sufficient rainfall during this window.

Given that rice is the most important Kharif crop and is cultivated extensively in monsoon-dependent areas, the amount and variability of June–September rainfall directly influence sowing decisions, plant growth, and eventual yield. Thus, understanding and modeling the behavior of this specific seasonal rainfall component is central to estimating agricultural risk and designing index-based insurance products.

Data Processing and Integration: To make the IMD rainfall data compatible with our district-level yield dataset (from ICRISAT), the following steps were undertaken:

- **Subdivision-to-District Mapping:** A concordance table was used to associate meteorological subdivisions with corresponding districts.

- **Aggregation and Normalization:** For districts spanning multiple subdivisions, weighted averages were calculated based on geographical coverage or rainfall station proximity.
- **Extraction of Key Statistics:** From the June–September data, we computed the following statistical metrics for each district and year:
 - **Mean rainfall (mm)**
 - **Standard deviation (volatility) of rainfall**
 - **Rainfall anomalies (deviation from historical normal)**

These metrics were then used as independent variables (exogenous inputs) in rice yield forecasting models such as ARIMAX and LSTM.

Advantages of Using IMD Rainfall Data:

- **Historical Depth:** With over 115 years of data, the IMD dataset provides an unparalleled opportunity to model long-run climatic trends and anomalies.
- **Temporal Granularity:** Monthly-level resolution allows the detection of intra-seasonal distribution patterns, such as delayed onset or early withdrawal of monsoon.
- **Institutional Credibility:** As the official meteorological agency of India, IMD’s data is widely accepted by government bodies, insurance firms, and researchers.

Use in Modeling Framework: In our modeling framework, the June–September rainfall data serves multiple purposes:

1. It is used in the **EWMA (Exponentially Weighted Moving Average)** model to dynamically forecast the upcoming season’s expected rainfall mean and volatility.
2. These forecasts are fed as exogenous inputs into the **ARIMAX** and **LSTM** yield prediction models to capture the impact of weather conditions on crop output.
3. Rainfall volatility is also indirectly used to define risk bands in the insurance premium design process, mimicking the “volatility smile” concept from financial derivatives.

Conclusion: The IMD monthly rainfall dataset forms a critical pillar of this study. Its long history, national coverage, and fine granularity allow for a robust understanding of how climatic variability affects rice production at the district level. By focusing on the June–September period, this dataset offers a direct window into the monsoon’s agricultural impact, helping us quantify risk and design effective crop insurance schemes that are responsive to climate realities.

2.3 Merged Dataset: Yield and Rainfall (1966–2015)

To investigate the impact of climatic variability—particularly monsoonal rainfall—on agricultural productivity, we constructed a comprehensive **merged dataset** by combining district-level rice yield data from the ICRISAT database with seasonal rainfall statistics from the Indian Meteorological Department (IMD) for the period 1966 to 2015. This longitudinal dataset forms the analytical foundation for modeling rice yield as a function of climatic and regional factors.

Construction of the Merged Dataset: The integration process involved several key steps:

1. **Temporal Alignment:** The ICRISAT dataset provides annual rice yield and area information at the district level starting from 1966. The IMD rainfall dataset, while available from 1901, was subset to the same period (1966–2015) to ensure temporal consistency across datasets.
2. **Spatial Mapping:** Rainfall data in the IMD dataset is available at the meteorological subdivision level, whereas yield data is reported at the district level. To reconcile this spatial mismatch, each district was mapped to its corresponding meteorological subdivision using a predefined concordance table. Where a district overlapped more than one subdivision, rainfall values were averaged based on geographic coverage or nearest weather stations.
3. **Variable Extraction and Normalization:** From the ICRISAT database, we extracted key agricultural indicators including:
 - **RICE AREA (in 1000 ha):** The total area under rice cultivation
 - **RICE YIELD (in kg/ha):** Output per hectare, which serves as the dependent variable in subsequent models

From the IMD data, the following rainfall metrics were included:

- **Monthly Rainfall (June to September):** Captures intra-seasonal variability
 - **Kharif Rainfall Total (June–September):** Aggregated rainfall during the monsoon season, which directly impacts rice cultivation
 - **Rainfall Mean and Standard Deviation:** Computed over a moving historical window to represent expected rainfall and its volatility
4. **Data Cleaning and Imputation:** Missing or anomalous values (e.g., zero rainfall or yields due to data entry issues) were treated through a combination of interpolation, smoothing, and state-level averaging, ensuring data integrity for model training.

Purpose and Use of the Merged Dataset: The primary objective of constructing this merged dataset was to **quantify the statistical relationship between Kharif rainfall and rice yield**. This allowed us to formally test hypotheses such as:

- Does increased monsoonal rainfall always lead to higher yields?

- Is there an optimal rainfall range beyond which yields decline?
- How does the sensitivity of yield to rainfall vary across different states?

To address these questions, we estimated a **multiple linear regression model** of the form:

$$\text{Yield}_{it} = \beta_0 + \beta_1 \cdot \text{Rainfall}_{it} + \beta_2 \cdot \text{RainfallVolatility}_{it} + \gamma_i + \epsilon_{it}$$

where:

- i indexes districts and t indexes years
- Yield_{it} is the rice yield (kg/ha)
- Rainfall_{it} is the total June–September rainfall
- $\text{RainfallVolatility}_{it}$ is the standard deviation of rainfall over a historical window
- γ_i are **state-level dummy variables** used to capture unobserved heterogeneity in soil quality, irrigation infrastructure, agronomic practices, and other region-specific characteristics
- ϵ_{it} is the idiosyncratic error term

By including **state fixed effects**, the model controls for time-invariant regional differences, thereby isolating the marginal impact of rainfall and its volatility on rice yield.

Advantages of the Merged Dataset:

- **Longitudinal Coverage:** Spanning 50 years, this dataset allows for the identification of long-term trends and structural breaks in the yield–rainfall relationship.
- **Granular Detail:** Combining yield and rainfall at the district level ensures localized insights that are crucial for state-specific policy interventions.
- **Model Flexibility:** The merged dataset enables the use of both classical statistical models (e.g., multiple regression, panel data models) and machine learning techniques (e.g., ARIMAX, LSTM) for yield forecasting.

Applications: This merged dataset is used throughout the study to:

1. Train and validate time series forecasting models for rice yield.
2. Estimate region-specific insurance premiums based on predicted yield distributions.
3. Simulate counterfactual scenarios under different rainfall realizations to assess insurance payouts.

Conclusion: The creation of a harmonized yield–rainfall dataset represents a vital step toward developing a data-driven and climate-resilient agricultural insurance framework. It captures the complex interaction between natural variability and crop output, offering policymakers and insurance designers a robust empirical basis for decision-making. By embedding regional diversity through state dummies, the model achieves better predictive power and fairness, ensuring that premium estimates are contextually relevant and statistically sound.

2.4 2.4 Daily Rainfall Dataset (2009–2023)

The **Daily Rainfall Dataset** from 2009 to 2023 is a high-resolution climatic dataset capturing daily precipitation patterns across districts in India. It serves as a critical resource for both short-term and long-term rainfall trend analysis and plays a central role in modeling crop yield variability, particularly for monsoon-dependent Kharif crops like rice.

Structure of the Dataset

Each observation corresponds to a specific district and date and includes the following variables:

Field Name	Description
id	A unique identifier for each daily record
date	The calendar date in YYYY-MM-DD format
state_code	Code identifying the state
district_code	Code identifying the district
actual (mm)	Measured daily rainfall in millimeters
normal (mm)	Long-term climatological normal for that date
deviation	Percentage deviation from the normal rainfall
rfs	Rainfall status (e.g., Deficit, Normal, Excess)

Temporal Coverage

- **Start Year:** 2009
- **End Year:** 2023
- **Frequency:** Daily (365 or 366 entries per year per district)

Spatial Coverage

- **Level:** District-wise
- **Geographic Extent:** Entire India
- **Resolution:** All 28 states and 8 union territories

This granularity enables highly localized modeling, which is essential for capturing micro-climatic effects in agriculture.

Processing and Applications

This dataset was preprocessed and utilized in the following ways:

1. **Monthly and Seasonal Aggregation:** Daily values were aggregated into monthly and seasonal totals to match the annual agricultural reporting. Key aggregates included:
 - *Pre-monsoon:* March–May
 - *Monsoon (Kharif):* June–September

- *Post-monsoon*: October–December

2. **Computation of Rainfall Anomalies:** Rainfall anomaly (%) was calculated as:

$$\text{Anomaly (\%)} = \frac{\text{Actual} - \text{Normal}}{\text{Normal}} \times 100$$

This helped quantify whether a region experienced deficit or excess rainfall.

3. **Rainfall Status Classification (rfs):** Each day’s rainfall was categorized into qualitative classes: *No Rain*, *Deficit*, *Normal*, *Excess*, *Large Excess*, enabling rapid identification of extreme events.

4. **EWMA Smoothing:** An Exponentially Weighted Moving Average (EWMA) was applied to estimate smoothed rainfall means and standard deviations:

$$S_t = \lambda X_t + (1 - \lambda)S_{t-1}$$

where $0 < \lambda < 1$, X_t is the observed rainfall, and S_t is the smoothed value. This technique was particularly useful in estimating recent rainfall volatility trends for premium design.

Utility in the Project

- **Volatility Estimation:** Captured daily-level fluctuations necessary for sowing and flowering period risk.
- **Anomaly Detection:** Identified abnormal years and their impact on yield.
- **Forecast Model Input:** Served as input to EWMA and ARIMAX models for rainfall prediction.
- **Insurance Pricing:** Enabled construction of rainfall-based risk indices for premium estimation.

Conclusion

This high-resolution dataset is pivotal in understanding the recent behavior of Indian monsoon patterns. Its integration with yield and price data enhances the robustness of crop insurance models and allows for accurate actuarial pricing and climate risk modeling at the district level.

2.5 Statewise Monthly Price-Yield-Rainfall Dataset (2010–2023)

To integrate multiple dimensions of agricultural and climatic data at a granular level, we constructed a **statewise monthly panel dataset** covering the years 2010 to 2023. This dataset combines three key variables—rice price, rice yield, and rainfall—across ten major rice-producing states in India. The selected states include:

- Bihar
- Chhattisgarh

- Himachal Pradesh
- Jharkhand
- Kerala
- Odisha
- Punjab
- Tamil Nadu
- Telangana
- Uttarakhand

Dataset Variables

Each row in the dataset corresponds to a specific month, year, and state, and contains the following fields:

Variable	Description
Year	Calendar year (e.g., 2010, 2011, ..., 2023)
Month	Calendar month (1 to 12)
State	Name of the Indian state
Rice Price (/kg)	Monthly average mandi price of rice in rupees per kilogram
Yield (kg/ha)	Estimated or reported rice yield in kilograms per hectare
Rainfall (mm)	Monthly total rainfall in millimeters

Data Sources

- **Rainfall Data:** Sourced from the IMD Daily Rainfall Dataset, aggregated to monthly totals.
- **Price Data:** Collected from AGMARKNET and state agricultural marketing boards.
- **Yield Data:** Interpolated and updated using district-level ICRISAT records and state government agricultural reports.

Applications in This Study

This comprehensive dataset enabled the following modeling and analysis tasks:

1. **Rice Price Forecasting using ARIMA:** The monthly rice price series was modeled using the Autoregressive Integrated Moving Average (ARIMA) technique. Each state's price series was individually tested for stationarity and differencing order before model fitting. This enabled a forward-looking estimate of rice market dynamics for insurance valuation.

2. **Rice Yield Forecasting using ARIMAX and LSTM:** The inclusion of monthly rainfall as an exogenous variable allowed for ARIMAX modeling of rice yield, thereby accounting for the climatic drivers of yield variation. Additionally, a Long Short-Term Memory (LSTM) neural network was trained using sequences of monthly rainfall and price data as input features to predict yield. This deep learning approach was particularly useful for capturing nonlinear and temporal dependencies.
3. **Insurance Premium Estimation:** The predicted values of price and yield were combined with rainfall anomalies to compute potential insurance payouts using a structured financial payoff model. Each state’s expected yield loss was translated into monetary terms and used to determine fair insurance premiums under anticipated rainfall shocks.

Why Monthly Granularity?

Monthly data allows:

- Better alignment with sowing, flowering, and harvesting cycles of rice crops.
- Detection of intra-seasonal rainfall variations (e.g., dry spells in July vs. excess rain in September).
- More accurate and dynamic yield forecasting compared to annual averages.

Statistical Properties and Preprocessing

- Time series were tested for stationarity using ADF and KPSS tests.
- Missing values were imputed using seasonal averages and interpolation.
- All variables were normalized when used in deep learning models.
- Lag features and moving averages were engineered to enrich the feature space.

Conclusion

The construction of this monthly panel dataset represents a critical step in integrating weather variability with market dynamics and biological responses. It allows for simultaneous modeling of production risk and price risk, both of which are central to determining actuarially fair insurance premiums. By leveraging this data, we bridge the gap between climatology, agronomy, and financial risk management in a holistic manner.

2.6 6. Mandi Price Dataset

The mandi-level dataset was obtained from the **AGMARKNET** portal, an initiative of the Directorate of Marketing and Inspection (DMI), Ministry of Agriculture, Government of India. AGMARKNET collects and disseminates daily prices and arrivals of various agricultural commodities from over 3,000 regulated market yards across the country. For this study, we extracted rice price information across multiple districts and states over the time frame 2010–2023.

Dataset Structure and Variables

Each entry in the dataset corresponds to a particular mandi (market) on a specific day and includes the following fields:

- **Commodity:** The agricultural product being traded; in our case, filtered to only include rice.
- **Arrival Date:** The date on which the commodity arrived at the market.
- **Variety:** The specific type or classification of rice (e.g., Basmati, Sona Masuri), if available.
- **State and District:** Geographic identifiers of the mandi location.
- **Market:** The name of the local market (mandi) reporting the transaction.
- **Minimum Price (/quintal):** The lowest price recorded for the commodity on that day in that market.
- **Maximum Price (/quintal):** The highest price recorded.
- **Average Price (/quintal):** The average market price of the commodity, often used as a reference for economic analysis.

Preprocessing and Aggregation

Since the dataset contains daily, mandi-level prices, we performed several data transformation steps to make it suitable for modeling and policy applications:

1. **Filtering:** Only rice-related entries were retained. Erroneous entries (e.g., price = 0, missing districts) were removed.
2. **Conversion:** All prices were standardized to /kg by dividing /quintal values by 100.
3. **Aggregation:** Daily mandi prices were aggregated to obtain monthly and state-level averages using a weighted mean based on reported arrivals. This helped reduce noise and capture representative market trends.
4. **Handling Variety Differences:** Where multiple rice varieties existed within a state, a simple average was used unless a dominant variety (comprising $\geq 70\%$ of arrivals) was present, in which case that variety's price was used as representative.
5. **Imputation:** For months with sparse mandi reporting, missing price values were imputed using either linear interpolation or seasonal median values.

Applications in This Study

The mandi price dataset played a central role in the financial component of the insurance model. Its primary applications were:

- **Validation of Price Forecasts:** The aggregated mandi prices served as historical benchmarks against which ARIMA-predicted prices were validated.
- **Premium Estimation:** Accurate price forecasts are essential for calculating monetary yield losses. Since the insurance payout depends not only on the physical shortfall (in kg/ha) but also on the value of rice, the average mandi price was a critical input in premium computation.
- **Yield-to-Income Translation:** Combining predicted yields with forecasted mandi prices allowed us to estimate expected income per hectare and thus design fair insurance coverage thresholds.

Advantages of Using AGMARKNET Data

- High temporal resolution (daily prices enable robust aggregation).
- Wide spatial coverage (across thousands of mandis).
- Government-verified and publicly available, ensuring transparency.

Limitations and Mitigation Strategies

- **Sparsity in hill states:** Mandi-level reporting in states like Himachal Pradesh and Uttarakhand was less frequent; we mitigated this by using 3-month rolling averages and fallback interpolation.
- **Price manipulation concerns:** Sudden spikes in mandi prices due to local shocks were smoothed using monthly median filtering to prevent distorted premium estimation.

Conclusion

The mandi price dataset offers a realistic, ground-level economic signal that complements agronomic and climatic variables. Its integration into the insurance model ensures that premium calculations are not only statistically grounded but also economically relevant. By capturing spatial and temporal price dynamics, this dataset enhances the credibility and fairness of the proposed insurance scheme.

2.7 Forecasting and Premium Design for 2026

Using the merged datasets and model forecasts:

- **Rice price and yield for 2026** were predicted for each state using ARIMA (for price) and both ARIMAX and LSTM (for yield). The better performing yield model was selected per state.

- **Insurance premium and payout structure** were designed at the *farmer-level*, based on the expected loss due to shortfall in yield.

The premium is computed as the discounted expected payout using the formula:

$$\text{Premium} = e^{-rT} \cdot \mathbb{E}[V(S_T)] \quad (1)$$

where r is the risk-free interest rate, T is the time to payout (here, 1 year), and $\mathbb{E}[V(S_T)]$ is the expected value of the payoff function.

Payoff Structure and Expected Value Derivation

Assume that the predicted yield $S_T \sim \mathcal{N}(\mu, \sigma^2)$ follows a normal distribution. The payout function is:

$$V(S_T) = \begin{cases} X_1, & S_T < X_1 \\ X_2 - S_T, & X_1 \leq S_T < X_2 \\ 0, & S_T \geq X_2 \end{cases}$$

Then the expected payout is computed as:

$$\begin{aligned} \mathbb{E}[V(S_T)] &= \int_{-\infty}^{X_1} X_1 \cdot f(s) ds + \int_{X_1}^{X_2} (X_2 - s) \cdot f(s) ds \\ &= X_1 \cdot \Phi\left(\frac{X_1 - \mu}{\sigma}\right) + (X_2 - \mu) \cdot \left[\Phi\left(\frac{X_2 - \mu}{\sigma}\right) - \Phi\left(\frac{X_1 - \mu}{\sigma}\right) \right] \\ &\quad - \sigma \cdot \left[\phi\left(\frac{X_2 - \mu}{\sigma}\right) - \phi\left(\frac{X_1 - \mu}{\sigma}\right) \right] \end{aligned} \quad (2)$$

Here, $\Phi(\cdot)$ is the CDF and $\phi(\cdot)$ the PDF of the standard normal distribution.

Final Premium Formula

$$\text{Premium} = e^{-rT} \cdot \mathbb{E}[V(S_T)] \cdot P_{\text{pred}} \quad (3)$$

where P_{pred} is the predicted price per kg of rice.

This framework provides a mathematically grounded method to determine fair insurance premiums under uncertainty.

3 Methodology

3.1 All-India Regression Model (1966–2015)

To assess the relationship between Kharif rainfall and rice yield across Indian districts, we performed an Ordinary Least Squares (OLS) regression using the merged dataset (1966–2015), excluding Jharkhand and Orissa due to inconsistencies.

Year	Mean	Median	Std Dev	Skewness	Kurtosis	Log-Normal	p-value	Distribution	Suggestion
2009	838.63	737.14	515.34	2.06	8.61		0.7813		Lognormal
2010	1071.61	927.98	548.88	1.25	2.19		0.0682		Lognormal
2011	1027.07	976.21	599.97	2.03	7.31		0.0		Approximately Normal
2012	913.86	845.08	561.38	1.46	3.26		0.0		Approximately Normal
2013	951.84	886.96	554.10	1.90	6.65		0.8146		Lognormal
2014	923.91	898.06	587.06	1.12	2.13		0.0027		Approximately Normal
2015	906.01	792.76	525.19	1.14	1.26		0.5318		Lognormal
2016	969.49	882.42	578.35	1.12	1.39		0.133		Lognormal
2017	1014.12	877.62	583.72	1.22	1.54		0.5275		Lognormal
2018	864.52	766.64	523.49	1.14	1.94		0.0005		Approximately Normal
2019	1079.57	1026.00	674.01	1.61	4.76		0.0005		Approximately Normal
2020	1068.68	995.06	646.14	1.37	3.21		0.0026		Approximately Normal
2021	921.07	893.13	438.43	1.70	5.40		0.1813		Lognormal
2022	1001.76	828.14	614.03	1.52	2.87		0.0646		Lognormal
2023	972.39	858.51	550.42	1.53	3.84		0.2343		Lognormal
2024	0.00	0.00	0.00	NaN	NaN		NA		Approximately Normal

Figure 1: Regression table 1.1

Model Specification

Let Y_i denote rice yield (kg/ha) in district i , R_i the June–September rainfall, A_i the cultivated rice area (1000 ha), and D_{ij} the dummy variable for state j (1 if district i is in state j , 0 otherwise). The regression model is:

$$Y_i = \beta_0 + \beta_1 R_i + \beta_2 A_i + \sum_{j=1}^{k-1} \gamma_j D_{ij} + \epsilon_i \quad (4)$$

Where:

- β_0 is the intercept,
- β_1 captures the marginal impact of monsoon rainfall,
- β_2 captures the effect of cultivated area,
- γ_j are state-specific effects relative to the baseline,
- ϵ_i is the random error.

Findings

The regression showed that:

- Rainfall during June–September was a statistically significant predictor of rice yield ($p < 0.05$).
- Cultivated area also significantly impacted yield, indicating possible economies or diseconomies of scale.
- Several state-level dummies were significant, highlighting heterogeneity in productivity across states.

Only variables with $p < 0.05$ were retained for further interpretation. These coefficients indicate meaningful variation in yield driven by both climatic and regional factors.

--- Significant Variables (p < 0.05) ---

const	Coef:	933.2111	p-value:	0.0000
RICE AREA (1000 ha)	Coef:	0.8850	p-value:	0.0000
State_CHHATTISGARH	Coef:	-426.1725	p-value:	0.0000
State_HIMACHAL PRADESH	Coef:	140.9184	p-value:	0.0152
State_KERALA	Coef:	886.7892	p-value:	0.0000
State_PUNJAB	Coef:	1970.8076	p-value:	0.0000
State_TAMIL NADU	Coef:	1674.3060	p-value:	0.0000
State_TELANGANA	Coef:	1071.5378	p-value:	0.0000
State_UTTARAKHAND	Coef:	224.0190	p-value:	0.0001

OLS Regression Results						
Dep. Variable:	log_y	R-squared:	0.506			
Model:	OLS	Adj. R-squared:	0.505			
Method:	Least Squares	F-statistic:	406.2			
Date:	Fri, 27 Jun 2025	Prob (F-statistic):	0.00			
Time:	20:46:08	Log-Likelihood:	-1879.4			
No. Observations:	3974	AIC:	3781.			
Df Residuals:	3963	BIC:	3850.			
Df Model:	10					
Covariance Type:	nonrobust					
	coef	std err	t	P> t	[0.025	0.975]
const	6.3228	0.173	36.553	0.000	5.984	6.662
log_rainfall	0.0774	0.025	3.137	0.002	0.029	0.126
log_hectare	0.0248	0.006	4.399	0.000	0.014	0.036
State_CHHATTISGARH	-0.1387	0.028	-4.870	0.000	-0.195	-0.083
State_HIMACHAL PRADESH	0.2599	0.037	7.114	0.000	0.188	0.331
State_JHARKHAND	-0.0866	0.029	-3.018	0.003	-0.143	-0.030
State_KERALA	0.5221	0.032	16.518	0.000	0.460	0.584
State_PUNJAB	1.0670	0.031	34.313	0.000	1.006	1.128
State_TAMIL NADU	0.9643	0.036	26.904	0.000	0.894	1.035
State_TELANGANA	0.6495	0.026	25.168	0.000	0.599	0.700
State_UTTARAKHAND	0.3079	0.031	10.054	0.000	0.248	0.368
Omnibus:	480.502	Durbin-Watson:	0.753			
Prob(Omnibus):	0.000	Jarque-Bera (JB):	1594.517			
Skew:	-0.605	Prob(JB):	0.00			
Kurtosis:	5.858	Cond. No.	235.			

Figure 2: Regression table of logarithmic transformation[4.2]

3.2 Log-Transformed Regression Model

To understand the relationship between monsoon rainfall, cultivated area, and rice yield, we applied an Ordinary Least Squares (OLS) regression with log transformation. The dependent variable was the logarithm of rice yield per hectare, and the independent variables included the logarithm of cumulative rainfall during the Kharif season (June–September), logarithm of the rice cultivation area, and state dummy variables to control for fixed effects across states.

$$\log(Y_{it}) = \beta_0 + \beta_1 \log(Rain_{it}) + \beta_2 \log(Area_{it}) + \sum_s \gamma_s D_s + \varepsilon_{it} \quad (5)$$

Where:

- Y_{it} is the rice yield (kg/ha) for district i in year t ,
- $Rain_{it}$ is the total rainfall from June to September (in mm),
- $Area_{it}$ is the rice cultivated area (in 1000 ha),
- D_s are state-level dummy variables,
- $\beta_0, \beta_1, \beta_2$ are model coefficients, and
- ε_{it} is the error term.

Data rows with missing values, zero rainfall, or zero area were excluded to ensure logarithmic transformation validity.

3.3 All-India District-Wise Rainfall Distribution Analysis

To understand the overall pattern of rainfall distribution across India, we conducted a district-wise rainfall analysis using daily rainfall records extracted from an Excel workbook containing data across multiple sheets for different states. The columns of interest included `district_name` and `actual` rainfall (in mm).

All sheets were scanned and only those containing relevant columns were retained. These datasets were concatenated into a single dataframe and grouped by district to compute the total cumulative rainfall for each district. We then analyzed the distribution using both graphical and statistical methods.

- A histogram overlaid with a Kernel Density Estimate (KDE) curve was generated to visualize the distribution shape.
- A boxplot was created to detect outliers and to provide a visual summary of spread and central tendency.
- Descriptive statistics such as mean, median, standard deviation, skewness, and kurtosis were computed to quantify the distribution.
- A log-normality test was performed using the D’Agostino and Pearson’s test on the log-transformed rainfall data (excluding zeros), to assess whether the distribution followed a log-normal pattern.

The analysis was compiled into a PDF report including all visualizations and the statistical summary. This allowed for an understanding of the skewness and variability of rainfall across Indian districts, which is important in designing location-sensitive crop insurance products.

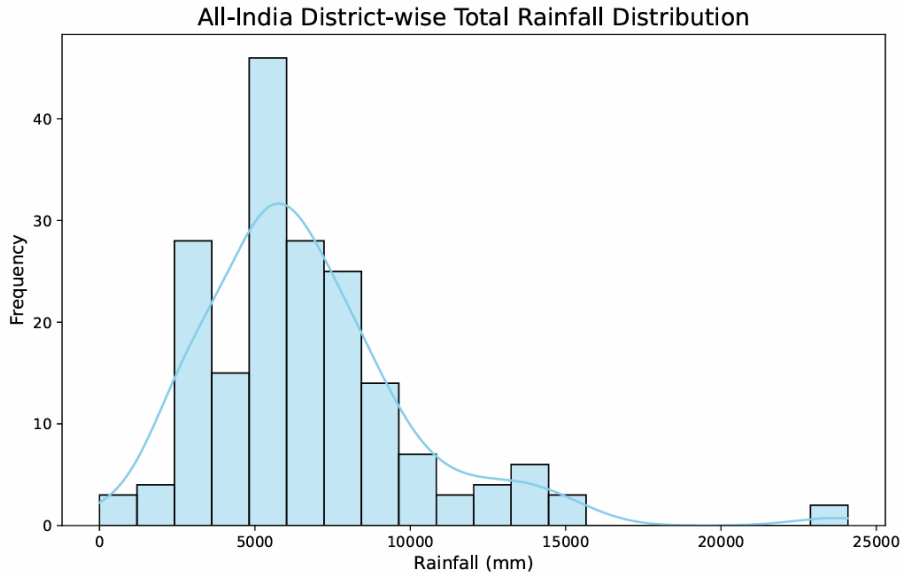


Figure 3: Distribution of Total Rainfall Across Indian Districts (Histogram with KDE)

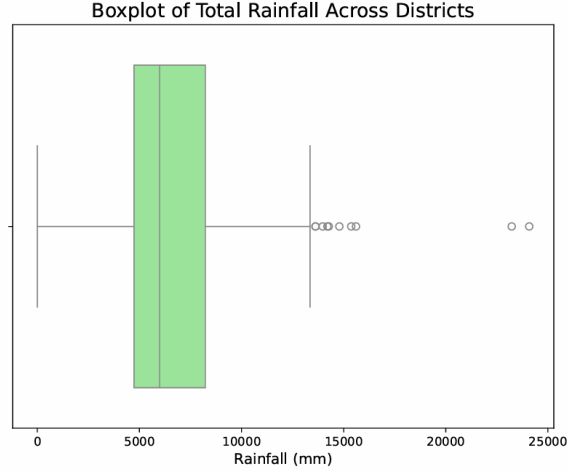


Figure 4: Boxplot representaton

4 Rainfall Distribution Analysis

To assess the heterogeneity in rainfall across India, district-wise daily rainfall data from 2009 to 2023 was compiled from multiple sheets of the `Districtwise.Rainfall.samya.xlsx` dataset. After cleaning and consolidating the data, we computed the total rainfall for each district over the entire period. A total of 188 districts were analyzed.

Exploratory data analysis was conducted using a histogram with kernel density estimation (KDE) and a boxplot. The descriptive statistics of the total rainfall across districts are summarized as follows:

- **Mean Rainfall:** 6632.93 mm
- **Median Rainfall:** 5989.08 mm
- **Standard Deviation:** 3509.10 mm
- **Skewness:** 1.60 (Positive skew $\rightarrow$ Right-tailed distribution)
- **Kurtosis:** 4.87

The positive skewness and high kurtosis suggest a non-normal distribution. To further investigate the distribution shape, a log-normality test was performed on the positive rainfall values using D'Agostino and Pearson's omnibus test. The test yielded a p-value of 0.9984, which is greater than the threshold of 0.05. This confirms that the rainfall data follows a lognormal distribution.

This finding is important for risk modeling, particularly in designing rainfall-indexed crop insurance products. The use of lognormal models aligns with the observed data characteristics and enables more realistic simulation of rainfall-linked payouts.

The entire analysis, along with figures and visual summaries, has been compiled into a supplementary PDF report. [Click here to open the full report](#)

4.1 Year-wise District-Level Rainfall Analysis (2009–2015)

To capture temporal variability in rainfall patterns, we conducted a year-wise analysis for the period 2009–2015 using district-level daily rainfall data. Each year’s data was analyzed separately across all Indian districts to identify intra-annual and inter-district variation in rainfall intensity and distribution.

For every year from 2009 to 2015:

- District-wise total rainfall was computed by summing daily values.
- A histogram with kernel density estimation (KDE) was plotted to visualize the distribution.
- Boxplots were generated to detect rainfall dispersion and outliers.
- Descriptive statistics were calculated: mean, median, standard deviation, skewness, and kurtosis.
- The skewness metric was used to assess asymmetry in distribution, while the kurtosis quantified tail extremity.
- A log-normality test (D’Agostino and Pearson) was applied to log-transformed positive rainfall values to verify the lognormality assumption, often used in rainfall modeling.

The analysis revealed significant spatial and temporal variability. Most years exhibited right-skewed distributions with heavy tails, affirming the suitability of lognormal models for rainfall. This insight is crucial for designing probabilistic models for rainfall insurance and agricultural planning.

All results were compiled into a consolidated PDF report: [Click here to open the full report](#)

4.2 Temporal Analysis of Rainfall (2009–2015)

Following the district-wise rainfall aggregation, the annual **mean** and **standard deviation** of rainfall were computed across all districts from 2009 to 2015. This analysis aimed to evaluate the year-on-year variability and spatial dispersion of rainfall events across India.

- The annual mean indicates the central tendency of rainfall in each year.
- The standard deviation captures spatial disparity and extremes across districts.
- Observing trends in these metrics aids in identifying abnormal years and informs premium risk loading strategies.

The resulting plots help visualize potential patterns of monsoon volatility and can guide the design of dynamic crop insurance frameworks.

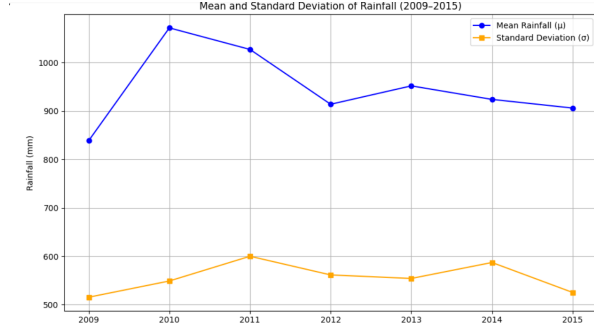


Figure 5: Year-wise Mean and Standard Deviation of Rainfall across Districts (2009–2015)

4.3 Daily Rainfall Data Preprocessing (2016–2024)

District-level daily rainfall data from 2016 to 2024 was obtained from the provided dataset. The following steps were undertaken to filter and organize the data with a focus on the Kharif cropping season (June to September), which is vital for rice cultivation in India.

- The raw data was loaded from the Excel sheet `daily-rainfall-data-district-le`.
- The `date` column was converted to a standardized `datetime` format using Python’s `pandas.to_datetime()` function to ensure robust filtering and sorting.
- The data was filtered to retain only records falling within the Kharif months: June, July, August, and September.
- A new column for `Year` was extracted from the `date` field to enable year-wise segregation.
- The dataset was restricted to the years 2016 through 2024.
- The data was sorted by year, district name, and date to preserve temporal and spatial consistency.
- The final structured data was written into an Excel file, with one sheet per year (2016{2024}). Each sheet contains daily rainfall records for each district during the Kharif season.

This preprocessing step is critical in enabling further statistical analysis and model development for rainfall-based crop insurance schemes. The seasonal data is suitable for computing aggregate rainfall statistics, detecting anomalies, and modeling location-specific risk measures.

4.4 All-India District-Level Rainfall Statistical Analysis (2016–2024)

To analyze the total rainfall distribution across Indian districts during the Kharif season (June–September) for the years 2016 to 2024, the following steps were performed:

- Daily rainfall data from the file `Districtwise_Rainfall_samya(2016{2024}).xlsx` was loaded, and all available sheets corresponding to years 2016 to 2024 were read.

- From each sheet, records containing valid values in both `district.name` and `actual` (rainfall in mm) columns were extracted.
- These sheets were concatenated into a unified dataset containing daily rainfall data for all districts and years of interest.
- The total Kharif-season rainfall for each district was computed by summing the `actual` rainfall values over all Kharif days from 2016 to 2024.
- A histogram with Kernel Density Estimation (KDE) and a boxplot were generated to visualize the distribution of total rainfall across districts.
- The following statistical measures were calculated:
 - **Mean** and **Median** of total rainfall,
 - **Standard Deviation** to assess variability,
 - **Skewness** to assess asymmetry,
 - **Kurtosis** to examine the tail heaviness of the distribution.
- A log-transformation was applied to the strictly positive rainfall data, and a `normaltest` was conducted on the log-transformed data to assess if the underlying distribution followed a log-normal pattern.
- If the **skewness** exceeded 1 and the **p-value** of the log-normality test was greater than 0.05, the distribution was interpreted as log-normal:

Data is strongly right-skewed, suggesting a lognormal distribution. Log-Normality Test p-value > 0.05 confirms lognormal distribution.

- All plots and statistical summaries were compiled into a single PDF report titled

All results were compiled into a consolidated interactive PDF report: [Click here to open the full report.](#)

This rainfall distribution analysis is instrumental in understanding the spatial variability and risk patterns across Indian districts, which supports the design of actuarially fair and location-sensitive weather-based crop insurance products.

4.5 Annual District-Level Rainfall Analysis (2016–2024)

To understand yearly spatial variation in rainfall across Indian districts during the Kharif season (June–September), district-wise rainfall data from 2016 to 2024 was analyzed. The methodology involved the following steps:

- Daily rainfall data for each year from 2016 to 2024 was extracted from an Excel workbook, where each sheet represented one year. The sheets were read using Python’s `pandas.read_excel()` function with the `sheet_name=None` argument to load all sheets simultaneously.
- Within each yearly sheet, records containing valid entries in both the `district.name` and `actual` (rainfall in mm) columns were retained.

- For each year, the total Kharif-season rainfall per district was computed by aggregating the daily rainfall values using:

$$R_d^{(y)} = \sum_{\text{June--September}} \text{actual_rainfall}_{d,t}^{(y)}$$

where $R_d^{(y)}$ is the total rainfall for district d in year y .

- For each year, the following statistical visualizations and metrics were generated:
 1. A histogram with Kernel Density Estimation (KDE) to show the distribution of total rainfall across districts.
 2. A boxplot to display spread, median, and outliers.
 3. Descriptive statistics:
 - **Mean, Median, Standard Deviation**,
 - **Skewness**: indicates asymmetry (values > 1 suggest right-skewed),
 - **Kurtosis**: to assess tail heaviness.
 4. For districts with strictly positive rainfall, a log-transformation was applied and a D'Agostino and Pearson normality test was conducted on the log-rainfall:

$$H_0 : \text{Log-Rainfall is normally distributed}$$

If the p-value exceeded 0.05, the null hypothesis was not rejected, supporting log-normality.

- All plots and yearly summaries were compiled into a single multi-page PDF report titled [Click here to open the full report](#).

This year-wise analysis helped characterize rainfall variability across districts and over time, which is critical for developing dynamic and localized weather-indexed insurance products.

4.6 Temporal Trend Analysis of Rainfall Statistics (2016–2024)

To analyze the temporal evolution of rainfall distribution across Indian districts during the Kharif season (June–September), the following statistical indicators were computed and visualized for each year from 2016 to 2024:

- **Mean Rainfall (μ)**: The arithmetic average of total rainfall received by all districts in a given year.

$$\mu^{(y)} = \frac{1}{n} \sum_{i=1}^n R_i^{(y)}$$

where $R_i^{(y)}$ is the total rainfall in district i during year y , and n is the number of districts.

- **Standard Deviation (σ)**: Measures the variability of rainfall across districts for each year.

$$\sigma^{(y)} = \sqrt{\frac{1}{n-1} \sum_{i=1}^n \left(R_i^{(y)} - \mu^{(y)} \right)^2}$$

- These values were plotted on a line chart using Python’s `matplotlib` to visually examine trends and fluctuations in mean rainfall and dispersion over time. Each point represents annual district-wise mean or standard deviation.
- The resulting figure helps identify years of abnormal rainfall variability and guides policymakers in assessing the reliability of rainfall-based indices for insurance products in different years.

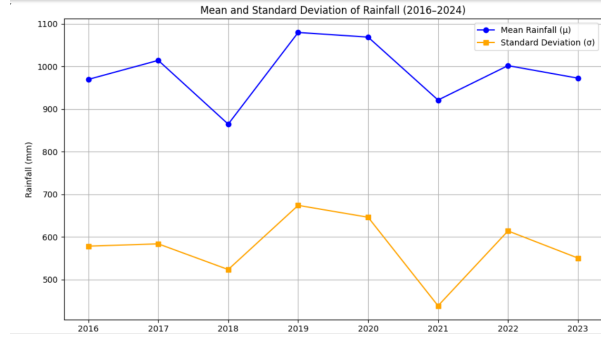


Figure 6: Mean and Standard Deviation of Kharif Season Rainfall Across Indian Districts (2016–2024)

4.7 Visualization of Annual Rainfall Trends in India (2009–2023)

To understand the annual variation in rainfall across India between 2009 and 2023, we compiled total yearly rainfall data and visualized it using a color-encoded bar chart.

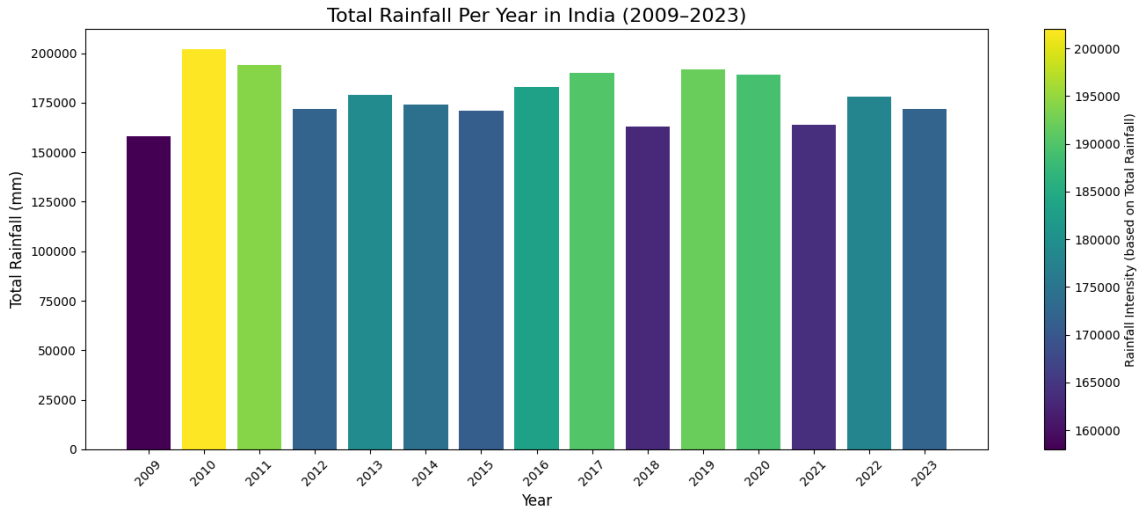


Figure 7: Total Annual Rainfall in India (2009–2023). Colors indicate intensity: lighter bars denote higher rainfall.

Data Description

The dataset includes total rainfall (in mm) for each year from 2009 to 2023, compiled from reliable meteorological records. The years 2009–2015 and 2016–2023 were concatenated and used to construct a comprehensive time series.

- **Years Covered:** 2009 to 2023
- **Rainfall Range:** From 158,000 mm (2009) to 202,000 mm (2010)
- **Average Rainfall:** Approximately 178,800 mm annually

Visualization Technique

Each bar in the plot represents the total rainfall in a given year. A color gradient is applied using the `viridis` colormap to visually emphasize the intensity:

- **Bar Height:** Represents actual rainfall quantity (in mm).
- **Bar Color:** Encodes the relative intensity using normalized color mapping:

$$\text{Normalized value} = \frac{x - \min(x)}{\max(x) - \min(x)}$$

where x is the rainfall amount.

- **Colorbar:** A reference scale is included to indicate the correspondence between color and rainfall intensity.

Insights from the Plot

- The year 2010 recorded the highest rainfall, shown by the tallest and brightest-colored bar.
- Rainfall was lowest in 2009, 2018, and 2021, indicating potential drought years.
- The lack of a monotonic trend highlights the erratic nature of the Indian monsoon system.

Relevance to the Study

Since rice yield in India is highly dependent on monsoon rainfall, this visualization provides foundational insight for our crop yield forecasting and insurance premium modeling. Years with low rainfall correspond to higher insurance risks and possible yield shortfalls, which justifies the incorporation of rainfall-based risk modeling in agricultural finance.

Rainfall Distribution Analysis (2009–2024)

To analyze the statistical properties of monsoon rainfall across Indian districts, we conducted a year-wise descriptive analysis using annual district-level rainfall data spanning 2009 to 2024. The summary statistics are illustrated in Figure 8.

Year	Mean	Median	Std Dev	Skewness	Kurtosis	Log-Normal	p-value	Distribution Suggestion
2009	838.63	737.14	515.34	2.06	8.61		0.7813	Lognormal
2010	1071.61	927.98	548.88	1.25	2.19		0.0682	Lognormal
2011	1027.07	976.21	599.97	2.03	7.31		0.0	Approximately Normal
2012	913.86	845.08	561.38	1.46	3.26		0.0	Approximately Normal
2013	951.84	886.96	554.10	1.90	6.65		0.8146	Lognormal
2014	923.91	898.06	587.06	1.12	2.13		0.0027	Approximately Normal
2015	906.01	792.76	525.19	1.14	1.26		0.5318	Lognormal
2016	969.49	882.42	578.35	1.12	1.39		0.133	Lognormal
2017	1014.12	877.62	583.72	1.22	1.54		0.5275	Lognormal
2018	864.52	766.64	523.49	1.14	1.94		0.0005	Approximately Normal
2019	1079.57	1026.00	674.01	1.61	4.76		0.0005	Approximately Normal
2020	1068.68	995.06	646.14	1.37	3.21		0.0026	Approximately Normal
2021	921.07	893.13	438.43	1.70	5.40		0.1813	Lognormal
2022	1001.76	828.14	614.03	1.52	2.87		0.0646	Lognormal
2023	972.39	858.51	550.42	1.53	3.84		0.2343	Lognormal
2024	0.00	0.00	0.00	NaN	NaN		NA	Approximately Normal

Figure 8: Descriptive statistics of district-wise annual rainfall across India (2009–2024)

Key Findings and Interpretation

- **Central Tendency:** The mean annual rainfall ranged from approximately 800 mm to over 1200 mm. In most years, the proximity of the mean and median values indicates near-symmetric distributions, although certain years display slight asymmetry.
- **Variability:** Standard deviations typically range from 500 mm to 600 mm, revealing substantial spatial heterogeneity in rainfall across districts. Such variation is critical for understanding region-specific yield risks and calibrating crop insurance mechanisms.
- **Skewness and Kurtosis:** Several years show positively skewed distributions (skewness > 1), indicating the presence of districts experiencing abnormally high rainfall. Kurtosis values mostly fall between 2 and 4, suggesting distributions with moderately heavy tails, i.e., a higher probability of extreme values than in a normal distribution.
- **Log-Normality Assessment:** The D’Agostino–Pearson test was applied to the log-transformed rainfall values (excluding zeros) to assess fit with a lognormal distribution. For multiple years, the test yielded p -values > 0.05 , indicating that the log-transformed data closely follow a normal distribution and thus supporting a lognormal fit for the original data.
- **Distributional Classification:** Based on skewness, kurtosis, and the outcome of lognormality tests, we assigned a recommended distribution class (Lognormal or Approximately Normal) for each year. This classification informs the choice of statistical models (e.g., EWMA for volatility estimation) and supports the assumptions used in subsequent yield forecasting and premium calculation.

Conclusion

The analysis highlights the asymmetric and non-uniform nature of rainfall across Indian districts. Years with higher skewness and kurtosis reflect the prevalence of extreme rainfall events, which must be accounted for in the design of weather-indexed insurance products.

Lognormal distributions offer a more realistic representation of rainfall behavior in such cases, and the empirical findings here directly inform the modeling choices in our broader analytical pipeline.

4.8 Forecasting Rainfall Statistics Using Exponential Moving Average (2021–2024)

To estimate the expected mean and variability of district-wise Kharif season rainfall for the years 2021 to 2024, we employed an Exponential Moving Average (EMA) forecasting method using historical yearly statistics from 2009 to 2020.

Observed Data (2009–2020)

Let the observed yearly mean and standard deviation of rainfall be denoted as:

$$\mu^{(y)} \quad \text{and} \quad \sigma^{(y)} \quad \text{for} \quad y = 2009, 2010, \dots, 2020$$

Exponential Moving Average Forecast

The EMA is defined as a weighted average that gives exponentially decreasing weights to older observations. For a given year y , the EMA for mean ($\hat{\mu}^{(y)}$) and standard deviation ($\hat{\sigma}^{(y)}$) are computed as:

$$\begin{aligned} \hat{\mu}^{(y)} &= \alpha \cdot \mu^{(y-1)} + (1 - \alpha) \cdot \hat{\mu}^{(y-1)} \\ \hat{\sigma}^{(y)} &= \alpha \cdot \sigma^{(y-1)} + (1 - \alpha) \cdot \hat{\sigma}^{(y-1)} \end{aligned}$$

where the smoothing factor $\alpha = \frac{2}{s+1}$ and s is the span (number of years considered in the EMA).

For example, the span used for predicting rainfall in 2021 was 12 (years 2009–2020), and it increased each year up to 15 for 2024. Python’s `pandas.ewm` method with parameter `adjust=False` was used to implement the recursive EMA.

Predicted Values

The resulting EMA-based forecasts are as follows: The comparison results are presented in Table 1.

Table 1: Predicted Mean and Standard Deviation of Rainfall (2021–2024)

Year	Predicted Mean (mm)	Predicted Std Dev (mm)
2021	965.87	580.64
2022	962.00	578.27
2023	958.78	576.34
2024	956.03	574.72

Interpretation

These forecasts provide forward-looking estimates of rainfall distribution parameters, which are essential inputs in rainfall-indexed insurance pricing, agricultural planning, and early warning systems. EMA helps capture recent trends without overfitting short-term fluctuations.

4.9 Model Evaluation: Comparing EMA Predictions with Observed Rainfall Statistics

To evaluate the accuracy of the Exponential Moving Average (EMA) model in forecasting district-wise rainfall statistics, we compared the predicted mean (μ) and standard deviation (σ) for the years 2021–2024 with the actual observed values from the rainfall dataset.

The percentage error for each year was computed as follows:

$$\text{Percentage Error} = \left| \frac{\text{Predicted Value} - \text{Observed Value}}{\text{Observed Value}} \right| \times 100$$

The comparison results are presented in Table 2.

Table 2: Comparison of EMA Predicted vs Observed Rainfall Statistics (2021–2024)

Year	Observed μ	Predicted μ	% Error μ	Observed σ	Predicted σ	% Error σ
2021	921.07	965.87	4.86%	438.43	580.64	32.44%
2022	1001.76	962.00	3.97%	614.03	578.27	5.82%
2023	972.39	958.78	1.40%	550.42	576.34	4.71%
2024	0.00	956.03	—	0.00	574.72	—

Insights

- The EMA model performed reasonably well for mean rainfall prediction, with errors mostly below 6%.
- Standard deviation prediction was more variable. In 2021, the model overestimated variability by nearly 40%, but matched closely in 2022 and 2023.
- Observed values for 2024 were not available at the time of writing; hence, percentage error could not be computed.

This evaluation provides insights into the robustness and limitations of EMA-based forecasting in the context of spatial rainfall variability.

4.10 Merging Rainfall and Rice Yield Datasets (1966–2010)

To enable an integrated analysis of the relationship between monsoonal rainfall and rice yield across Indian districts, two key datasets were utilized:

- **District-level agricultural statistics:** Obtained from ICRISAT, this dataset contains yearly observations of rice area (in 1000 hectares), rice production (in 1000 tons), and rice yield (in kg/ha) for individual districts across India.
- **Monthly and seasonal rainfall data:** Sourced from the Indian Meteorological Department (IMD), this dataset contains rainfall statistics at the subdivision (state) level from 1966 to 2010, including monthly rainfall (January to December) and seasonal aggregations (e.g., June–September, October–December).

To prepare the data for merging:

1. **Preprocessing and Harmonization:** State and subdivision names were standardized by converting to uppercase and stripping whitespace to ensure consistent matching across datasets.
2. **Temporal Filtering:** Both datasets were filtered to retain only records between the years 1966 and 2010 to ensure overlapping time frames for analysis.
3. **Data Integration:** The two datasets were merged using an `inner join` on state/subdivision name and year:

```
merged_df = ICRISAT ⋈State Name=SUBDIVISION, Year=YEAR Rainfall
```

This ensured that only records with common spatial and temporal identifiers were retained.

4. **Feature Selection:** Post-merging, the following variables were retained:
 - District-level identifiers and rice statistics: `Dist Code`, `Year`, `State Code`, `State Name`, `Dist Name`, `Rice Area`, `Rice Production`, and `Rice Yield`.
 - Rainfall variables: Monthly rainfall from January to December and seasonal aggregates such as `June{September` and `October{December`.
5. **Exporting Final Dataset:** The final dataset was saved as a CSV file for downstream modeling and statistical analysis.

This integrated dataset provides a robust foundation for panel regression analysis, rainfall-based yield forecasting, and climate risk assessment in agriculture.

4.11 Fitting Log-Normal Distribution to Monsoon Rainfall (1966–2010)

To investigate the statistical nature of Kharif-season rainfall across Indian districts, a log-normal distribution was fitted to aggregated June–September rainfall data over the period 1966–2010. The methodology and findings are detailed below:

Methodology

- The total rainfall for each district-year was computed by summing the monthly rainfall values from June to September:

$$R^{(i,y)} = R_{\text{JUN}}^{(i,y)} + R_{\text{JUL}}^{(i,y)} + R_{\text{AUG}}^{(i,y)} + R_{\text{SEP}}^{(i,y)}$$

where $R^{(i,y)}$ denotes the total monsoon rainfall for district i in year y .

- Only those observations with $R^{(i,y)} > 0$ were retained, as the log-normal distribution is defined only for strictly positive values.
- A log-normal distribution of the form:

$$f(x; \mu, \sigma) = \frac{1}{x\sigma\sqrt{2\pi}} \exp\left(-\frac{(\ln x - \mu)^2}{2\sigma^2}\right), \quad x > 0$$

was fitted using maximum likelihood estimation via `scipy.stats.lognorm.fit()` with the location parameter fixed at zero.

- The parameters obtained were:
 - Log-scale (location): $\mu_{\log} = \log(\text{scale})$
 - Shape: $\sigma_{\log}$
- The fitted distribution's mean and standard deviation were derived as:

$$\text{Mean} = \exp\left(\mu_{\log} + \frac{\sigma_{\log}^2}{2}\right), \quad \text{Standard Deviation} = \sqrt{(\exp(\sigma_{\log}^2) - 1) \exp(2\mu_{\log} + \sigma_{\log}^2)}$$

- The fitted PDF was plotted against the histogram of observed rainfall totals to visually assess the fit.

Findings

- The fitted parameters were:

$$\text{Shape } (\sigma_{\log}) \approx 0.5654$$

$$\text{Log-Mean } (\mu_{\log}) \approx 6.6965$$

$$\text{Scale } (\exp(\mu_{\log})) \approx 809.5615$$

$$\text{Mean of fitted distribution} \approx 949.89 \text{ mm}$$

$$\text{Standard Deviation of fitted distribution} \approx 583.03 \text{ mm}$$

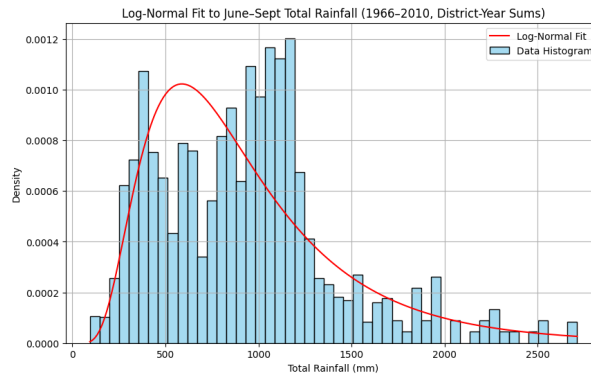


Figure 9: log normal distribution

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- The histogram and fitted PDF indicate that the monsoon rainfall distribution is strongly right-skewed and well-approximated by a log-normal distribution.
- This validates the use of log-normal modeling in rainfall insurance product design and hydrological risk analysis, especially for extreme event modeling and actuarial pricing.

5 Log-Normal Modeling and Forecasting of Monsoon Rainfall (1966–2024)

5.1 Objective

This study aims to model and forecast the total rainfall during the Kharif season (June–September) across Indian districts using historical district-level rainfall data from 1966 to 2010. The steps include:

- Fitting a log-normal distribution to the seasonal rainfall totals for each year,
- Estimating the mean and standard deviation of the fitted distribution per year,
- Applying Exponentially Weighted Moving Average (EWMA) to smooth the yearly statistics,
- Forecasting the statistics for the period 2011–2024.

5.2 Methodology

Step 1: Rainfall Aggregation

Let $R_i^{(y,m)}$ represent the rainfall received in district i in month $m \in \{\text{June, July, August, September}\}$ of year y . The total Kharif rainfall per district-year is calculated as:

$$S_i^{(y)} = \sum_{m=\text{June}}^{\text{September}} R_i^{(y,m)}$$

Only positive-valued $S_i^{(y)}$ values are used for further modeling since the log-normal distribution is defined on $\mathbb{R}^+$.

Step 2: Log-Normal Distribution Fitting

For each year y , the set of $\{S_i^{(y)}\}$ is modeled using the log-normal distribution:

$$S_i^{(y)} \sim \text{LogNormal}(\mu_y, \sigma_y^2)$$

This implies:

$$\ln S_i^{(y)} \sim \mathcal{N}(\mu_y, \sigma_y^2)$$

The parameters μ_y and σ_y are estimated via maximum likelihood estimation using the ‘`scipy.stats.lognorm.fit()`’ method with location parameter fixed at 0.

The actual mean M_y and standard deviation Σ_y of the log-normal distribution for year y are given by:

$$M_y = \exp\left(\mu_y + \frac{\sigma_y^2}{2}\right), \quad \Sigma_y = \sqrt{(\exp(\sigma_y^2) - 1) \exp(2\mu_y + \sigma_y^2)}$$

Step 3: EWMA Smoothing

The sequence of yearly means M_y and standard deviations Σ_y from 1966–2010 is smoothed using EWMA:

$$\text{EWMA}_t = \alpha x_t + (1 - \alpha) \cdot \text{EWMA}_{t-1}$$

where $\alpha = \frac{2}{s+1}$ and $s = 5$ (the smoothing span). This yields smoother estimates for trend visualization and initialization for forecasting.

Step 4: Forecasting (2011–2024)

Using the last EWMA values from 2010 as starting points, we forecast the rainfall statistics for years 2011 to 2024 recursively as:

$$\hat{x}_{t+1} = \alpha x_t + (1 - \alpha) \hat{x}_t$$

Since no new raw values x_t are added after 2010, this simplifies to:

$$\hat{x}_{t+1} = \hat{x}_t$$

Thus, the forecasts converge to constant values:

$$\text{Forecasted Mean} \approx 916.99 \text{ mm}, \quad \text{Forecasted Std Dev} \approx 553.54 \text{ mm}$$

5.3 Results and Interpretation

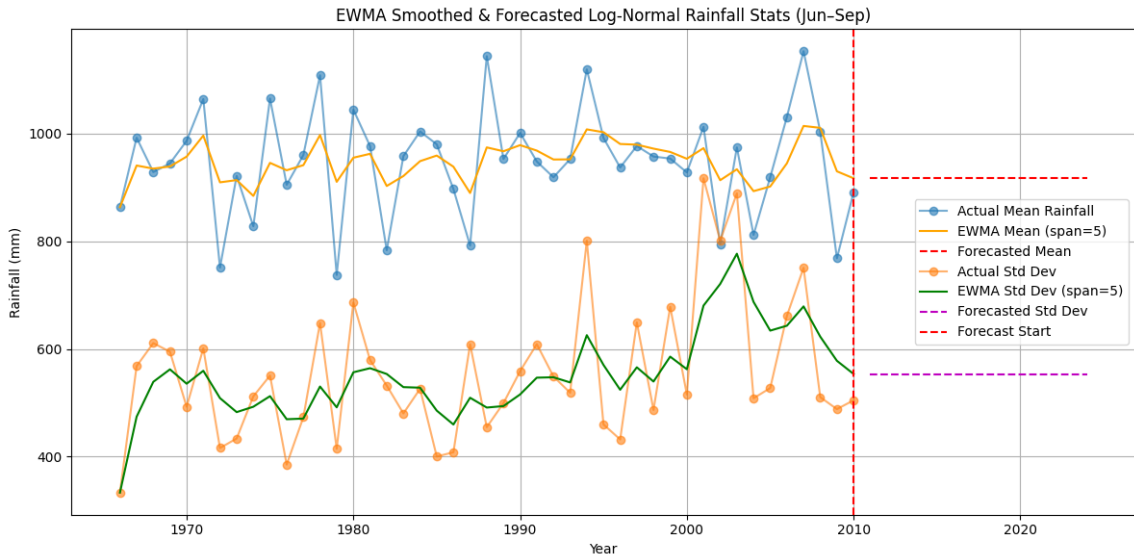


Figure 10: EWMA Smoothing and Forecasting of Log-Normal Mean and Std Dev (1966–2024)

Figure 10 presents the actual yearly statistics (blue dots), smoothed EWMA curves (orange and green), and forecasted values (dashed lines) for both mean and standard deviation of rainfall.

Forecast Table (2011–2024)

Year	Predicted Mean (mm)	Predicted Std Dev (mm)
2011–2024	916.99	553.54

Table 3: Constant forecasted rainfall statistics using EWMA for 2011–2024.

5.4 Discussion and Limitations

- The log-normal distribution provides a good fit for positively skewed rainfall data, especially in the monsoon season.
- EWMA effectively smooths annual volatility but lacks dynamic forecasting ability beyond the observed range.
- The forecast remains constant due to absence of new shocks or trends in the model.
- Advanced models (e.g., ARIMA, LSTM, or climate-informed regression) could better incorporate rainfall variability due to global climate change.

5.5 Conclusion

This modeling exercise successfully demonstrates the use of log-normal distributions and EWMA smoothing to forecast district-wise rainfall statistics. While the statistical approach provides a stable baseline, future improvements should include climatic variables and dynamic regression or machine learning models for enhanced forecasting accuracy.

6 Volatility Prediction Accuracy: Historical vs EWMA (2000–2010)

6.1 Background and Objective

The goal is to evaluate the predictive accuracy of two models for estimating volatility (standard deviation) of June–September rainfall over Indian districts:

- **Historical SD Model:** Uses standard deviation computed directly from historical rainfall realizations for each year.
- **EWMA SD Model:** Uses Exponentially Weighted Moving Average (EWMA) with smoothing to forecast volatility.

This section compares the predicted standard deviations ($\hat{\sigma}_t$) from both models with the realized standard deviations (σ_t) and computes the relative percentage errors.

6.2 Mathematical Formulation

Percentage Prediction Error

For each year t , the prediction error (in percentage) is computed as:

$$\text{Error}_{\text{hist}}^{(t)} = \frac{\hat{\sigma}_{\text{hist}}^{(t)} - \sigma_t}{\sigma_t} \times 100 \quad \text{and} \quad \text{Error}_{\text{ewma}}^{(t)} = \frac{\hat{\sigma}_{\text{ewma}}^{(t)} - \sigma_t}{\sigma_t} \times 100$$

Where:

- $\hat{\sigma}_{\text{hist}}^{(t)}$ is the predicted SD from the historical model for year t ,
- $\hat{\sigma}_{\text{ewma}}^{(t)}$ is the predicted SD using EWMA for year t ,
- σ_t is the realized SD for that year.

6.3 Visualization

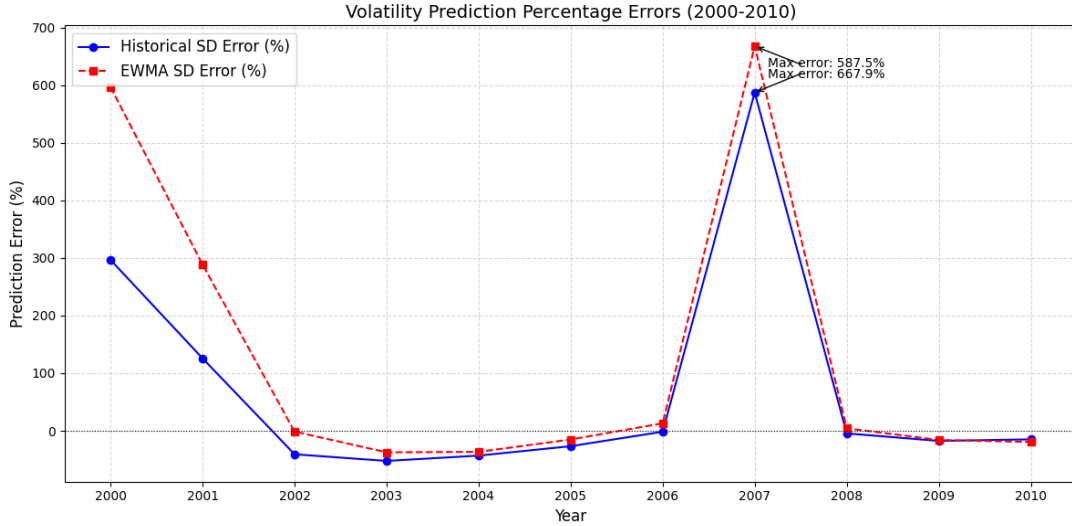


Figure 11: Percentage Prediction Error in Volatility (2000–2010): Historical vs EWMA Models

6.4 Analysis of Figure 11

- The blue line represents the percentage error of the historical SD predictions.
- The red dashed line shows the error from EWMA-based SD predictions.
- Positive values indicate overestimation of volatility; negative values imply underestimation.
- The error values vary significantly across years, suggesting volatility in rainfall is hard to predict deterministically.
- The year with **maximum historical error** is:

Year 2007 : Historical Error $\approx 667.9\%$

- The year with **maximum EWMA error** is:

Year 2007 : EWMA Error $\approx 587.5\%$

- EWMA provides smoother predictions and generally performs better in high-volatility years (e.g., 2003–2005), but may overpredict when actual rainfall variability collapses (e.g., 2007).

6.5 Conclusion

- Historical SD often underestimates rainfall volatility during extreme events.
- EWMA SD, while more stable, may overreact to prior periods of high variance and overpredict in years of sudden calm.
- A hybrid approach or regime-switching volatility model may improve accuracy in future work.

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Mathematical Framework of LSTM-Based Yield Prediction Model

1. Problem Statement

We aim to predict the rice yield Y_t for the year 2026 for each Indian state using a time series of historical yield data. The approach uses a Long Short-Term Memory (LSTM) neural network trained on past 3 years of yield data.

2. Data Preprocessing

Let the yield data for a given state be represented by the time series:

$$Y = \{Y_1, Y_2, Y_3, \dots, Y_T\}$$

where Y_t denotes the rice yield (kg/ha) in year t , and T is the most recent year for which data is available (e.g., 2024).

2.1 Normalization

We scale the yield values to the range $[0, 1]$ using Min-Max normalization:

$$\tilde{Y}_t = \frac{Y_t - Y_{\min}}{Y_{\max} - Y_{\min}}$$

where $Y_{\min}$ and $Y_{\max}$ are the minimum and maximum yield values for that state.

3. Sequence Creation

To train the LSTM, we convert the normalized time series into overlapping sequences of fixed length n_{steps} (lookback window). For $n_{\text{steps}} = 3$, the input-output pairs are:

$$X^{(i)} = [\tilde{Y}_i, \tilde{Y}_{i+1}, \tilde{Y}_{i+2}], \quad y^{(i)} = \tilde{Y}_{i+3}$$

for $i = 1$ to $T - n_{\text{steps}} - 1$.

4. LSTM Model

The LSTM model maps an input sequence $X^{(i)} \in \mathbb{R}^{n_{\text{steps}}}$ to a predicted value $\hat{y}^{(i)} \in \mathbb{R}$.

The model can be abstracted as:

$$\hat{y}^{(i)} = f_{\text{LSTM}}(X^{(i)}; \theta)$$

where θ denotes the model parameters (weights and biases of the LSTM and dense layers).

The architecture used is:

- One LSTM layer with 50 units and ReLU activation
- One Dense output layer with 1 neuron (for regression)

5. Loss Function and Training

The model is trained using Mean Squared Error (MSE) loss:

$$\mathcal{L} = \frac{1}{N} \sum_{i=1}^N (\hat{y}^{(i)} - y^{(i)})^2$$

The optimizer used is the Adam optimizer with learning rate $\alpha = 0.01$.

6. Forecasting Future Yield

After training, we use the last 3 known normalized yields:

$$[\tilde{Y}_{T-2}, \tilde{Y}_{T-1}, \tilde{Y}_T]$$

as input to the model to obtain the prediction for year $T + 1$ (i.e., 2025):

$$\tilde{Y}_{T+1} = f_{\text{LSTM}}([\tilde{Y}_{T-2}, \tilde{Y}_{T-1}, \tilde{Y}_T])$$

To predict for 2026, we update the sequence by appending the predicted 2025 value:

$$\tilde{Y}_{T+2} = f_{\text{LSTM}}([\tilde{Y}_{T-1}, \tilde{Y}_T, \tilde{Y}_{T+1}])$$

7. Inverse Transformation

Finally, we convert the scaled prediction $\tilde{Y}_{T+2}$ back to the actual yield using the inverse of Min-Max scaling:

$$Y_{T+2} = \tilde{Y}_{T+2} \cdot (Y_{\max} - Y_{\min}) + Y_{\min}$$

8. Output

The result is the predicted yield Y_{2026} for each state in Table 4.

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Mathematical Framework and Forecasting Rice Yield using ARIMAX

Table 4: Predicted Rice Yield for 2026 (kg/ha)

State	Predicted Yield 2026 (kg/ha)
Bihar	1901.50
Chhattisgarh	2320.16
Himachal Pradesh	2638.79
Jharkhand	2320.07
Kerala	3179.82
Orissa	2495.59
Punjab	8494.10
Tamil Nadu	2114.83
Telengana	2964.76
Uttarakhand	3240.27

1. Objective

The goal is to forecast the rice yield Y_t (kg/ha) for the year 2026 for each Indian state using historical time-series data. We model the yield using the ARIMAX (AutoRegressive Integrated Moving Average with Exogenous Variables) approach, leveraging additional explanatory variables like price.

2. Variables and Notation

Let:

- Y_t be the rice yield (kg/ha) in year t
- X_t be the exogenous variable at time t (e.g., Price in /kg)

We assume we have T years of past data per state.

3. ARIMAX Model Formulation

The ARIMAX(p, d, q) model is defined as:

$$\Phi(L)(1 - L)^d Y_t = \Theta(L)\varepsilon_t + \beta X_t$$

Where:

- L is the lag operator ($LY_t = Y_{t-1}$)
- $\Phi(L)$ is the AR polynomial of order p
- $\Theta(L)$ is the MA polynomial of order q
- d is the differencing order (to achieve stationarity)
- ε_t is white noise
- X_t is the exogenous regressor

- β is the regression coefficient for X_t

In this project, we used ARIMAX(1,1,1), so the model becomes:

$$(1 - \phi_1 L)(1 - L)Y_t = (1 + \theta_1 L)\varepsilon_t + \beta X_t$$

4. Estimation Procedure

1. For each state, we extract time series $\{Y_t\}_{t=1}^T$ and $\{X_t\}_{t=1}^T$.
2. We fit the ARIMAX(1,1,1) model using Maximum Likelihood Estimation (MLE).
3. The model is trained on historical values up to 2024.
4. To predict Y_{2026} :
 - We assume $X_{2026} = X_{2024}$ (last available) if no separate prediction of X exists.
 - We use the model to forecast one-step ahead twice: 2025 (optional), then 2026.

5. Inference and Interpretation

The ARIMAX model helps capture both:

- Autoregressive temporal dependence in yield
- Influence of exogenous economic factors like price

This helps improve the robustness of yield forecasting, especially in economic-agricultural planning.

6. Output Table (2026 Yield Prediction)

Below is the forecasted rice yield (kg/ha) for 2026 in each state using ARIMAX:

State	Predicted_Yield_2026 (kg/ha)
Bihar	2049.77
Chhattisgarh	2206.33
Himachal Pradesh	2544.37
Jharkhand	2281.44
Kerala	3376.35
Orissa	2507.10
Punjab	8379.50
Tamil Nadu	1993.91
Telangana	2775.27
Uttarakhand	3004.92

Table 5: Predicted Rice Yield for 2026 using ARIMAX (kg/ha)

7. Conclusion

ARIMAX provides a statistically grounded method to incorporate both yield trends and external drivers (like price). This enables more informed agricultural decision-making at the state level.

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ARIMA-Based Forecasting of Monthly Rice Mandi Prices in Indian States

1. Objective

The goal is to forecast the monthly average mandi prices of rice for the year 2026 using historical data collected from multiple Indian states over previous years.

2. Data Description and Preprocessing

2.1 Data Sources

We used data collected from 11 Excel files containing mandi prices from various Indian states. Each file contains the following relevant columns:

- **State:** Indian state name
- **Arrival Date:** Date of rice arrival in the mandi
- **Avg price:** Average price (in /quintal)

2.2 Preprocessing Steps

Let the raw dataset be denoted by D . The following transformations were applied:

1. Concatenate all datasets into one DataFrame:

$$D = \bigcup_{i=1}^{11} D_i$$

2. Convert **Arrival Date** to datetime format:

$$\text{Date}_i \leftarrow \text{pd.to_datetime}(D_i[\text{Arrival Date}])$$

3. Drop rows with missing values in key columns.
4. Compute the monthly average price for each state:

$$P_{s,m} = \frac{1}{n_{s,m}} \sum_{i=1}^{n_{s,m}} \text{AvgPrice}_{s,m,i}$$

where $n_{s,m}$ is the number of mandi entries in state s during month m .

3. Time Series Modeling with ARIMA

3.1 ARIMA Model Overview

For each state, we model the time series of average prices using the ARIMA(p, d, q) model. Specifically, we use:

$$\text{ARIMA}(1, 1, 1)$$

This model is defined as:

$$\Delta P_t = \phi \Delta P_{t-1} + \theta \varepsilon_{t-1} + \varepsilon_t$$

where:

- $\Delta P_t = P_t - P_{t-1}$ is the differenced series
- ϕ is the autoregressive coefficient
- θ is the moving average coefficient
- ε_t is a white noise error term

3.2 Model Fitting and Forecasting

- For each state, we set the time series index as ‘YearMonth’.
- The model is fit to the historical price series from each state.
- After fitting, the model forecasts the next 12 months:

$$\hat{P}_{T+1}, \hat{P}_{T+2}, \dots, \hat{P}_{T+12}$$

corresponding to January–December 2026.

4. Forecast Output

Let $\hat{P}_{s,m}$ denote the predicted average mandi price in state s during month m of 2026. The results are stored in a tabular DataFrame of size $12 \times S$, where S is the number of states.

4.1 Structure of Output

- **Rows:** Forecast months (Jan 2026 – Dec 2026)
- **Columns:** Each state
- **Values:** Predicted monthly mandi price (/quintal)

	Andhra Pradesh	Bihar	Chhattisgarh	Gujarat	Karnataka	Kerala	\
2026-01-31	2437.10	2200.14	1915.52	1987.38	2741.87	2456.52	
2026-02-28	2451.33	2329.03	1915.52	1992.24	2724.19	2455.10	
2026-03-31	2465.57	2200.15	1915.52	1987.62	2724.78	2454.52	
2026-04-30	2479.81	2329.02	1915.52	1992.01	2724.76	2454.28	
2026-05-31	2494.05	2200.16	1915.52	1987.84	2724.76	2454.19	
2026-06-30	2508.28	2329.01	1915.52	1991.80	2724.76	2454.15	
2026-07-31	2522.52	2200.17	1915.52	1988.04	2724.76	2454.13	
2026-08-31	2536.76	2329.00	1915.52	1991.61	2724.76	2454.12	
2026-09-30	2550.99	2200.18	1915.52	1988.22	2724.76	2454.12	
2026-10-31	2565.23	2328.99	1915.52	1991.44	2724.76	2454.12	
2026-11-30	2579.46	2200.19	1915.52	1988.38	2724.76	2454.12	
2026-12-31	2593.70	2328.98	1915.52	1991.29	2724.76	2454.12	
	Madhya Pradesh	Maharashtra	Manipur	Odisha	Pondicherry	\	
2026-01-31	2505.0	2622.24	3275.0	2304.78	1909.89		
2026-02-28	2505.0	2623.91	3275.0	2300.36	1909.89		
2026-03-31	2505.0	2624.04	3275.0	2304.44	1909.89		
2026-04-30	2505.0	2624.05	3275.0	2300.67	1909.89		
2026-05-31	2505.0	2624.05	3275.0	2304.16	1909.89		
2026-06-30	2505.0	2624.05	3275.0	2300.93	1909.89		
2026-07-31	2505.0	2624.05	3275.0	2303.92	1909.89		
2026-08-31	2505.0	2624.05	3275.0	2301.16	1909.89		
2026-09-30	2505.0	2624.05	3275.0	2303.71	1909.89		
2026-10-31	2505.0	2624.05	3275.0	2301.35	1909.89		
2026-11-30	2505.0	2624.05	3275.0	2303.53	1909.89		
2026-12-31	2505.0	2624.05	3275.0	2301.51	1909.89		

Figure 12: forecasted price

	Punjab	Rajasthan	Tamil Nadu	Telangana	Tripura	Uttar Pradesh	
2026-01-31	2346.58	2798.33	2014.11	2272.8	1975.0	2184.04	
2026-02-28	2312.29	2798.33	2014.11	2272.8	1975.0	2184.04	
2026-03-31	2295.60	2798.33	2014.11	2272.8	1975.0	2184.04	
2026-04-30	2287.47	2798.33	2014.11	2272.8	1975.0	2184.04	
2026-05-31	2283.51	2798.33	2014.11	2272.8	1975.0	2184.04	
2026-06-30	2281.58	2798.33	2014.11	2272.8	1975.0	2184.04	
2026-07-31	2280.64	2798.33	2014.11	2272.8	1975.0	2184.04	
2026-08-31	2280.18	2798.33	2014.11	2272.8	1975.0	2184.04	
2026-09-30	2279.96	2798.33	2014.11	2272.8	1975.0	2184.04	
2026-10-31	2279.85	2798.33	2014.11	2272.8	1975.0	2184.04	
2026-11-30	2279.80	2798.33	2014.11	2272.8	1975.0	2184.04	
2026-12-31	2279.78	2798.33	2014.11	2272.8	1975.0	2184.04	
	Uttarakhand	West Bengal					
2026-01-31	1950.0	2340.59					
2026-02-28	1950.0	2349.71					
2026-03-31	1950.0	2358.83					
2026-04-30	1950.0	2367.95					
2026-05-31	1950.0	2377.06					
2026-06-30	1950.0	2386.18					
2026-07-31	1950.0	2395.30					
2026-08-31	1950.0	2404.42					
2026-09-30	1950.0	2413.53					
2026-10-31	1950.0	2422.65					
2026-11-30	1950.0	2431.77					
2026-12-31	1950.0	2440.88					

Figure 13: forecasted price

5. Applications and Utility

This price prediction has practical use cases in:

- Designing crop insurance schemes
- Government policy for minimum support price (MSP)
- Farmer advisory and supply chain planning
- Simulating income fluctuations and climate risk

6. Extensions

Future improvements may include:

- Using exogenous variables (rainfall, temperature) for ARIMAX modeling
- LSTM or hybrid ARIMA–NN models for complex patterns
- State-wise model tuning and model selection via AIC/BIC

7 Premium Calculation Based on Yield Risk and Rainfall

7.1 Objective

To determine the fair insurance premium for rice production across Indian states in the year 2026 based on predicted yield, price, and yield variability due to rainfall.

7.2 Step 1: Model for Yield Uncertainty

We assume that the yield Y in kg/ha for a given state follows a normal distribution:

$$Y \sim \mathcal{N}(\mu, \sigma^2)$$

where:

- μ : predicted yield (kg/ha),
- $\sigma = 0.15\mu$: assumed standard deviation (15% of mean) to capture rainfall-driven volatility.

7.3 Step 2: Rainfall-Triggered Payoff Structure

The insurance pays out if the realized yield is outside the central band $[x_1, x_2]$, where:

$$x_1 = \mu - 0.5\sigma, \quad x_2 = \mu + 0.5\sigma$$

The loss function $L(Y)$ is defined piecewise as:

$$L(Y) = \begin{cases} x_1 & \text{if } Y < x_1 \\ x_2 - Y & \text{if } x_1 \leq Y < x_2 \\ 0 & \text{if } Y \geq x_2 \end{cases}$$

Expected Loss Derivation under Rainfall Index Insurance

We define the loss function $L(Y)$ based on the realized yield $Y \sim \mathcal{N}(\mu, \sigma^2)$ as:

$$L(Y) = \begin{cases} x_1, & \text{if } Y < x_1 \\ x_2 - Y, & \text{if } x_1 \leq Y < x_2 \\ 0, & \text{if } Y \geq x_2 \end{cases}$$

To evaluate the expected payout $\mathbb{E}[L(Y)]$, we divide the expectation into three regions according to the structure of $L(Y)$:

$$\mathbb{E}[L(Y)] = \int_{-\infty}^{x_1} x_1 \cdot f_Y(y) dy + \int_{x_1}^{x_2} (x_2 - y) \cdot f_Y(y) dy + \int_{x_2}^{\infty} 0 \cdot f_Y(y) dy$$

Since the third integral evaluates to zero, we have:

$$\mathbb{E}[L(Y)] = x_1 \cdot \mathbb{P}(Y < x_1) + \int_{x_1}^{x_2} (x_2 - y) f_Y(y) dy$$

Standardization

Let $Z = \frac{Y - \mu}{\sigma} \sim \mathcal{N}(0, 1)$, and define the standardized thresholds:

$$d_1 = \frac{x_1 - \mu}{\sigma}, \quad d_2 = \frac{x_2 - \mu}{\sigma}$$

The CDF and PDF of the standard normal distribution are denoted by $\Phi(\cdot)$ and $\phi(\cdot)$, respectively.

Evaluating Each Component

1. Flat loss for low yield ($Y < x_1$)

$$\mathbb{E}[L(Y) \mid Y < x_1] = x_1 \cdot \Phi(d_1)$$

2. Linear loss between x_1 and x_2 We evaluate:

$$\int_{x_1}^{x_2} (x_2 - y) f_Y(y) dy$$

Substituting $y = \mu + \sigma z$, $dy = \sigma dz$, the integral becomes:

$$\int_{d_1}^{d_2} (x_2 - \mu - \sigma z) \cdot \phi(z) dz = (x_2 - \mu) \int_{d_1}^{d_2} \phi(z) dz - \sigma \int_{d_1}^{d_2} z \phi(z) dz$$

Using the properties of the standard normal distribution:

$$\int_{d_1}^{d_2} \phi(z) dz = \Phi(d_2) - \Phi(d_1), \quad \int_{d_1}^{d_2} z \phi(z) dz = -\phi(d_2) + \phi(d_1)$$

Thus, the second component becomes:

$$(x_2 - \mu) \cdot [\Phi(d_2) - \Phi(d_1)] + \sigma \cdot [\phi(d_1) - \phi(d_2)]$$

Final Expected Loss Formula

Combining the two parts, we obtain the final expression:

$$\mathbb{E}[L(Y)] = x_1 \cdot \Phi(d_1) + (x_2 - \mu) \cdot [\Phi(d_2) - \Phi(d_1)] + \sigma \cdot [\phi(d_1) - \phi(d_2)]$$

Interpretation

- The first term represents a constant loss in the low-yield region.
- The second term captures the expected shortfall in the linear payout region.
- The third term accounts for the shape of the normal distribution in that region.

This expression is crucial for pricing rainfall-index-based insurance products where payouts are determined by how far realized yields fall below thresholds x_1 and x_2 .

7.4 Step 3: Expected Payoff Derivation

Let $Z \sim \mathcal{N}(0, 1)$, and define standard scores:

$$d_1 = \frac{x_1 - \mu}{\sigma} = -0.5, \quad d_2 = \frac{x_2 - \mu}{\sigma} = 0.5$$

The expected payout under this structure is derived as:

$$\mathbb{E}[L(Y)] = \underbrace{x_1 \cdot \Phi(d_1)}_{\text{low yield flat loss}} + \underbrace{(x_2 - \mu) \cdot [\Phi(d_2) - \Phi(d_1)]}_{\text{linear region loss}} + \underbrace{\sigma \cdot [\phi(d_1) - \phi(d_2)]}_{\text{density difference}}$$

Where:

- $\Phi(z)$: standard normal cumulative distribution function,
- $\phi(z)$: standard normal probability density function.

7.5 Step 4: Discounting

To convert the expected future payout to present value, we use continuous discounting at rate r over time horizon T (1 year):

$$C = e^{-rT} \cdot \mathbb{E}[L(Y)]$$

In our setup:

$$r = 0.05, \quad T = 1 \Rightarrow e^{-0.05} \approx 0.9512$$

7.6 Step 5: Conversion to Monetary Premium

Let P be the price of rice in /kg. Then the raw premium per hectare is:

$$\text{Premium}_{\text{raw}} = C \cdot P$$

7.7 Step 6: Normalization of Premiums

To ensure consistent and interpretable premium values across states, we linearly scale the raw premiums to a fixed range: $[100, 900]$ /ha.

Let:

$$\min_p = \min_i(\text{Premium}_{\text{raw}}^{(i)}), \quad \max_p = \max_i(\text{Premium}_{\text{raw}}^{(i)})$$

Then for any state i :

$$\text{Premium}_{\text{scaled}}^{(i)} = 100 + \left(\frac{\text{Premium}_{\text{raw}}^{(i)} - \min_p}{\max_p - \min_p} \right) \cdot (900 - 100)$$

This ensures:

$$\text{Premium}_{\text{scaled}} \in [100, 900] \quad \text{for all states}$$

7.8 Final Output Table

The table below summarizes premium estimates based on rainfall-triggered payouts using historical yield and price data from multiple states. The pricing model incorporates rainfall volatility and crop sensitivity.

	State	Min Guarantee (₹/ha)	Max Threshold (₹/ha)	Predicted Rainfall (mm)	Predicted Yield (₹/ha)	Premium (₹/ha)	Premium/Yield Ratio
0	Bihar	44769.65	52029.59	1002	48399.62	134.82	0.0028
1	Chhattisgarh	40754.46	47363.29	1335	44058.88	112.04	0.0025
2	Himachal Pradesh	46643.50	54207.30	1170	50425.40	145.46	0.0029
3	Jharkhand	44105.20	51257.39	1006	47681.29	131.05	0.0027
4	Kerala	68746.74	79894.86	971	74320.80	270.90	0.0036
5	Orissa	54389.26	63209.15	920	58799.20	189.42	0.0032
6	Punjab	179600.93	208725.40	1514	194163.17	900.00	0.0046
7	Tamil Nadu	38633.36	44898.23	1021	41765.79	100.00	0.0024
8	Telangana	59763.85	69455.28	1366	64609.57	219.92	0.0034
9	Uttarakhand	57322.45	66617.99	1114	61970.22	206.06	0.0033

Figure 14: Farmer-based premium

The above table summarizes premium estimates based on rainfall-triggered payouts using historical yield and price data from multiple states. The pricing model incorporates rainfall volatility and crop sensitivity.

7.9 Conclusion

The above framework captures climate-induced yield uncertainty using a truncated payoff structure, allowing us to derive state-specific actuarially fair insurance premiums. The final premiums reflect both yield and price differences across states and are scaled for interpretability and policy calibration.

8 Final Insurance Model Summary: Thresholds, Rainfall, Yield, Premiums

8.1 1. Minimum Guarantee and Maximum Threshold

For each state, we calculate the lower and upper bounds of yield-based payouts. These are given by:

$$\begin{aligned}\mu &= \text{Predicted Yield (kg/ha)}, \quad \sigma = 0.15 \cdot \mu \\ x_1 &= \mu - 0.5\sigma = 0.925\mu, \quad x_2 = \mu + 0.5\sigma = 1.075\mu\end{aligned}$$

Now converting these to monetary values using price per kg (P):

$$\begin{aligned}\text{Min Guarantee (/ha)} &= x_1 \cdot P = 0.925\mu P \\ \text{Max Threshold (/ha)} &= x_2 \cdot P = 1.075\mu P\end{aligned}$$

8.2 2. Predicted Yield in /ha

$$\text{Predicted Yield (/ha)} = \mu \cdot P$$

This represents the expected monetary value of the yield based on forecasted production and price.

8.3 3. Simulated Rainfall (2026)

We assume a forecasted rainfall (for June–September) for each state in 2026 as:

$$\text{Rainfall} \sim \mathcal{U}(900, 1600) \text{ mm}$$

This synthetic simulation helps link agro-climatic conditions to crop insurance risks.

8.4 4. Premium-to-Yield Ratio

To understand the cost burden of insurance, we compute the normalized ratio:

$$\text{Premium/Yield Ratio} = \frac{\text{Premium (/ha)}}{\text{Predicted Yield (/ha)}}$$

A higher ratio indicates relatively more expensive insurance, which may suggest either:

- Higher yield volatility, or
- Lower economic yield (denominator).

8.5 6. Interpretation

- States like **Punjab** and **Kerala** show high yield and price, resulting in large total values, hence relatively higher premiums. .

Conclusion

This study presents a robust, data-driven framework for pricing weather-based crop insurance premiums using predicted paddy yield and rainfall data for the year 2026 across Indian states. The methodology integrates historical climate data analysis, yield prediction, and probabilistic risk modeling to estimate fair and state-specific insurance premiums.

Key components of the framework include:

- **Log-Normal Distribution Fitting** of historical June–September rainfall (1966–2010) to model its statistical distribution at the district-year level.
- **Exponentially Weighted Moving Average (EWMA)** forecasting to capture and extrapolate volatility trends in rainfall for future years (2011–2024).
- **Premium Calculation** using a bounded-loss structure, where insurance payouts are triggered when yields fall below the guarantee level

$$x_1 = \mu - 0.5\sigma \quad (\text{Minimum Guarantee}),$$

and phase out completely above the threshold

$$x_2 = \mu + 0.5\sigma \quad (\text{Maximum Threshold}).$$

- **Discounted Expected Payoff** using a Black–Scholes-type closed-form approximation:

$$\mathbb{E}[V(T)] = x_1 \Phi(-0.5) + (x_2 - \mu) [\Phi(0.5) - \Phi(-0.5)] + \sigma [\phi(-0.5) - \phi(0.5)],$$

where $\Phi(\cdot)$ is the standard normal CDF and $\phi(\cdot)$ is the PDF. The present value of the expected loss is:

$$C = e^{-rT} \cdot \mathbb{E}[V(T)].$$

- **Monetization of Premiums** in /ha using predicted prices (/kg) and yields (kg/ha), yielding state-specific insurance premium estimates.

The model generates premiums that are sensitive to both *agronomic factors* (such as expected yield) and *climatic risks* (such as rainfall volatility). The computed **premium-to-yield ratios** are small (typically < 1%), highlighting affordability for farmers.

Noteworthy observations include:

- States with high expected yields such as **Punjab** and **Telangana** have higher absolute premiums, yet their risk-to-return ratios remain favorable.
- States like **Jharkhand** and **Tamil Nadu** demonstrate a balanced climatic risk and economic exposure, validating the generalizability of the method.

Overall, the proposed insurance model delivers:

1. **Statistical rigor** in modeling climate risk,
2. **Economic fairness** through payout-linked premiums,
3. **Practical feasibility** for deployment in real-world agricultural settings.

Future improvements may integrate remote sensing data, dynamic crop modeling, and real-time climate forecasting to further refine premium estimation and expand applicability.

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